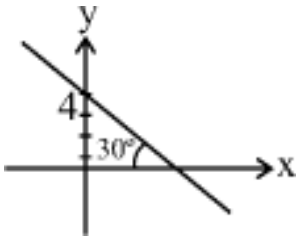
PHYSICS

1) 

Equation of given line is:

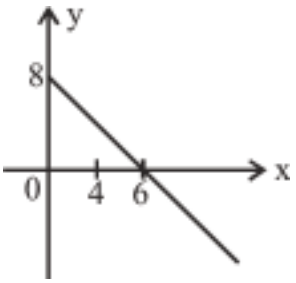
(1) $y = \sqrt{3}x + 4$

(2) $y = -\sqrt{3}x + 4$

(3) $y = \frac{1}{\sqrt{3}}x + 4$

(4) $y = -\frac{1}{\sqrt{3}}x + 4$

2) The equation of straight line shown in figure is :

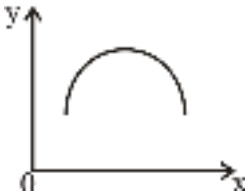


(1) 6x + 8y = 15

(2) 4x + 3y = 18

(3) 2y + 6x = 7

(4) 3y + 4x = 24

3) 

For the given curve

(1) slope is continuously decreasing

(2) slope is continuously increasing

(3) slope is constant

(4) slope is always positive

4) A dog is at point A(0, 3, 4)m and cat is at
B(5, 3, –8)m The dog is free to move but cat is fixed. The minimum distance travelled by dog to catch the cat is :-

(1) 25m

(2) 12m

(3) 13m

(4) 20m

5) The value of $\int_0^1 9x^8 dx + \int_0^{\pi/2} \cos x \, dx$ is :-

(1) 1

(2) 3

(3) 4

(4) 2

6) If $y = e^{\sin x}$, then $\frac{dy}{dx}$ = ?

(1) $-\cos x \cdot e^{\sin x}$

(2) $e^{\cos x}$

(3) $e^{-\cos x}$

(4) None of these

7) Value of $\int_{-\pi/2}^{\pi/2} \cos x dx$ is :-

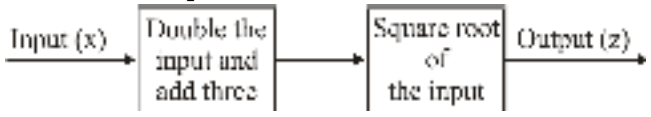
(1) Zero

(2) 1

(3) 2

(4) –2

8) In the given figure, each box represents a function machine. A function machine illustrates what it does with the input.



Which of the following statements is correct ?

(1) z = 2x + 3

(2) z = 2(x + 3)

(3) $z = \sqrt{2x+3}$

(4) $z = \sqrt{2(x+3)}$

9)

Find $\sqrt{1.04}$?

(1) 0.98
(2) 1
(3) 1.02
(4) 1.04

10)

The value of sin(23°) is :-

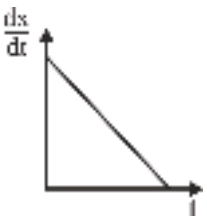
(1) $\frac{\sqrt{3}+1}{2\sqrt{2}}$
(2) $\frac{\sqrt{3}-1}{2\sqrt{2}}$
(3) $\frac{4\sqrt{3}+3}{10}$
(4) $\frac{4\sqrt{3}-3}{10}$

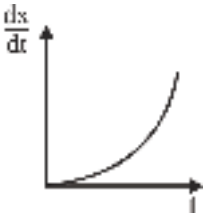
11) $\int \left(\sqrt{t}+\frac{1}{\sqrt{t}}\right)^2$ dt is :-

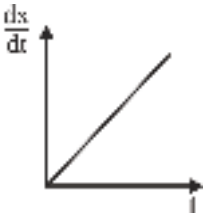
(1) $\frac{t^2}{2} + \log_e t + 2t + c$
(2) $t^2 + t + 2t + c$
(3) $2t^2 + \log_e t + 2t^2 + c$
(4) $\frac{t^2}{2} + \log t^2 + 2t + c$

12) If $x = t^2$, then graph between $\left(\frac{dx}{dt}\right)$ and t will be:-

(1) 

(2) 

(3) 

(4) 

13) Find the sum of series :
S = 7 + 10 + 13 + _ _ _ _ _ + 25

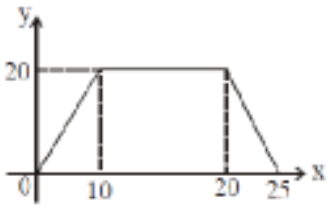
(1) 112
(2) 110
(3) 120
(4) 130

14) The radius of a circular plate increases at the rate of 0.1 cm/sec. At what rate does the area increase when the radius of plate in $\frac{5}{11}$ cm.

(1) 1 cm^2/sec
(2) 0.1 cm^2/sec
(3) 0.6 cm^2/sec
(4) 0.29 cm^2/sec

15) The velocity v of a particle is given by the equation $v = 6t^2 - 6t^3$ where v is in m/sec and t is time in seconds then :

(1) at t = 0, velocity is maximum
(2) at $t = \frac{2}{3}$, velocity is minimum
(3) minimum velocity is zero
(4) minimum velocity is – 2m/sec

16) Calculate $\int_0^{25} y.dx = ?$ 

(1) 150
(2) 250
(3) 350

(4) 450

17) y = (secx + tanx) (secx – tanx) :

(1) $\frac{dy}{dx} = \tan x + 1$

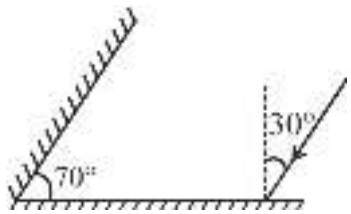
(2) $\frac{dy}{dx} = \tan x - 1$

(3) $\frac{dy}{dx} = \sec^2 x - \tan^2 x$

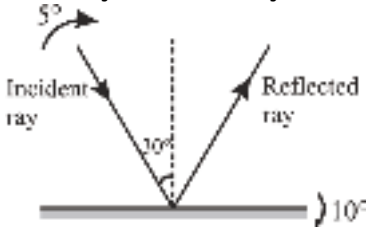
(4) $\frac{dy}{dx} = 0$

18) A ray of light incidents on a plane mirror at an angle of 30°. The deviation produced in the ray is :-

(1) 30°
(2) 60°
(3) 90°
(4) 120°

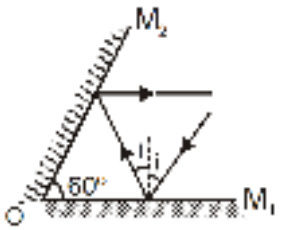
19) Find angle of deviation after two reflection :- 

(1) 120
(2) 240
(3) 220
(4) 115

20) Figure shows a plane mirror on which a light ray is incident. If incident ray is turned by 5° and the mirror by 10°, as shown, the angle turned by the reflected ray is :- 

(1) 15°, clockwise
(2) 25°, clockwise
(3) 30°, clockwise
(4) 25°, anticlockwise

21) Two plane mirrors are inclined to one another at an angle of 60°. A ray is incident on mirror M_1 at an angle i parallel to M_2. The reflected ray from mirror M_2 is parallel to mirror M_1 as shown in

figure. The angle of incidence i is 

(1) 20°
(2) 10°
(3) 30°
(4) 40°

22) Our eyes are most sensitive for which colour :

(1) Violet
(2) Blue
(3) Yellow
(4) Red

23) A vector having magnitude 30 units makes equal angles with x, y and z axes the components of vector along each of x, y, z axes are :-

(1) $10\sqrt{3}$ unit
(2) $\frac{10}{\sqrt{3}}$ unit
(3) $15\sqrt{3}$ units
(4) 10 unit

24) The magnitude of the x and y components of $\vec{P}$ are 7 and 6, and the magnitude of the x and y components of $\vec{P} + \vec{Q}$ are 11 and 9 respectively what is the magnitude of $\vec{Q}$ -

(1) 5
(2) 6
(3) 9
(4) 8

25) Resultant of two forces $\vec{F}_1$ and $\vec{F}_2$ has magnitude 50 N. The resultant is inclined to $\vec{F}_1$ at 60° and to $\vec{F}_2$ at 30°. Magnitudes of $\vec{F}_1$ and $\vec{F}_2$ respectively are :-

(1) 25 N, $25\sqrt{3}$N
(2) 20 N, $20\sqrt{3}$N
(3) 20 N, 30 N
(4) 30 N, 40 N

26) Which of the following group of concurrent forces may be in equilibrium (R = 0)?

	A	B	C
(1)	10	20	40
(2)	3	5	1
(3)	20	20	20
(4)	40	30	5

(1) 1

(2) 2

(3) 3

(4) 4

27) The velocity of a particle is $\vec{v} = 6\hat{i} + 2\hat{j} - 2\hat{k}$. The component of the velocity of a particle parallel to vector $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ in vector form is :-

(1) $6\hat{i} + 2\hat{j} + 2\hat{k}$

(2) $2\hat{i} + 2\hat{j} + 2\hat{k}$

(3) $\hat{i} + \hat{j} + \hat{k}$

(4) $6\hat{i} + 2\hat{j} - 2\hat{k}$

28) $\vec{A} = 2\hat{i} + \hat{j}, \vec{B} = 3\hat{j} - \hat{k}$ and $\vec{C} = 6\hat{i} - 2\hat{k}$ Value of $\vec{A} - 2\vec{B} + 3\vec{C}$ would be :-

(1) $20\hat{i} + 5\hat{j} + 4\hat{k}$

(2) $20\hat{i} - 5\hat{j} - 4\hat{k}$

(3) $4\hat{i} + 5\hat{j} + 20\hat{k}$

(4) $5\hat{i} + 4\hat{j} + 10\hat{k}$

29) If the sum of two unit vectors is a unit vector, then the magnitude of their difference is :-

(1) $\sqrt{2}$

(2) $\sqrt{3}$

(3) $1/\sqrt{2}$

(4) $\sqrt{5}$

30) If $\vec{A}$ and $\vec{B}$ are two non–zero vectors such that $|\vec{A} + \vec{B}| = \frac{|\vec{A} - \vec{B}|}{2}$ and $|\vec{A}| = 2|\vec{B}|$ then the angle between $\vec{A}$ and $\vec{B}$ is :

(1) 37°

(2) 53°

(3) $\cos^{-1}(-3/4)$

(4) $\cos^{-1}(-4/3)$

31) Two vectors $\vec{A}$ and $\vec{B}$ have equal magnitudes. If magnitude of $\vec{A} + \vec{B}$ is equal to 3 times the

magnitude of $\vec{A}-\vec{B}$, then angle between $\vec{A}$ and $\vec{B}$ is :

(1) 37°
(2) 53°
(3) 90°
(4) 0°

32) A vector $\vec{Q}$ which has a magniude of 8 is added to the vector $\vec{P}$ which lies along x-axis. The resultant of two vectors lies along y-axis and has magnitude twice that of $\vec{P}$. The magnitude of $\vec{P}$ is -

(1) $\frac{6}{\sqrt{5}}$
(2) $\frac{8}{\sqrt{5}}$
(3) $\frac{12}{\sqrt{5}}$
(4) $\frac{16}{\sqrt{5}}$

33)

If $\vec{A}=3\hat{i}+4\hat{j}+\hat{k}$ and $\vec{B}=\hat{i}+\hat{j}+\hat{k}$ are the diagonal of parallogram. Find area of parallogram.

(1) $\sqrt{14}$
(2) $\sqrt{\frac{7}{2}}$
(3) $\sqrt{7}$
(4) $\sqrt{\frac{2}{7}}$

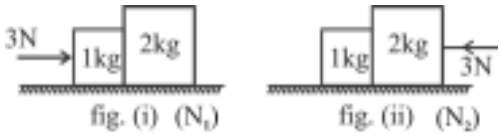
34) A body of mass 2 kg moving with a speed of 100 m/s hits perpendicular to wall and rebounds with the same speed. If the contact time is (1/50) s, the average force applied on the wall by the body is :-

(1) 10^4 N
(2) 2×10^4 N
(3) 4 N
(4) 8 N

35) A 600 kg rocket is for vertical firing. If the exhaust speed of gases is 1000 m/sec then the gas ejected per second to supply the thrust needed to overcome the gravitational force of the rocket is (g = 10 m/s^2):-

(1) 3 kg/sec
(2) 6 kg/sec
(3) 2 kg/sec

(4) 5 kg/sec

36) Two blocks of masses 1 kg and 2 kg are placed in contact on a smooth horizontal surface and a horizontal force 3N is applied on block 1 kg and an block 2 kg respectively in fig. (i) and fig. (ii). If normal force of interaction between both blocks are N_1 and N_2 according to both fig (i) and fig (ii) respectively. 

Then the value of $\frac{N_1}{N_2}$ is :-

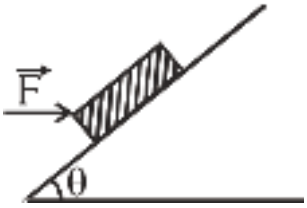
(1) $\frac{2}{1}$

(2) $\frac{1}{2}$

(3) $\frac{1}{4}$

(4) $\frac{1}{3}$

37) The figure shows a horizontal force $\vec{F}$ acting on a block of mass M on an inclined plane (angle θ).

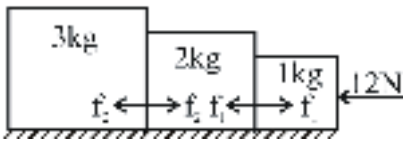
What is the normal reaction on the block ? 

(1) mgsinθ + Fcosθ

(2) mgsinθ – Fcosθ

(3) mgcosθ – Fsinθ

(4) mgcosθ + Fsinθ

38) Contact forces f_1 and f_2 are :- 

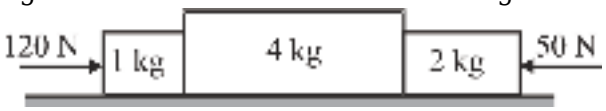
(1) 10N, 6N

(2) 6N, 10N

(3) 3N, 2N

(4) 2N, 3N

39) Three blocks of mass 1 kg, 4kg and 2kg are placed on a smooth horizontal plane as shown in the figure. The contact force between 1kg block and 4 kg block is



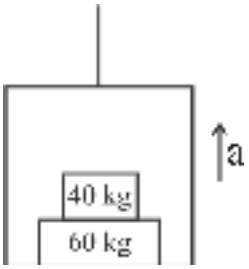
(1) 100 N

(2) 120 N

(3) 50 N

(4) 110 N

40) On the floor of an elevator, a block of mass 60 kg is placed on which another block of mass 40 kg is also placed. The elevator is moving up with a constant acceleration $5 m/s^2$. Force exerted by 40 kg block on the 60 kg block is : ($g = 10\ m/s^2$)



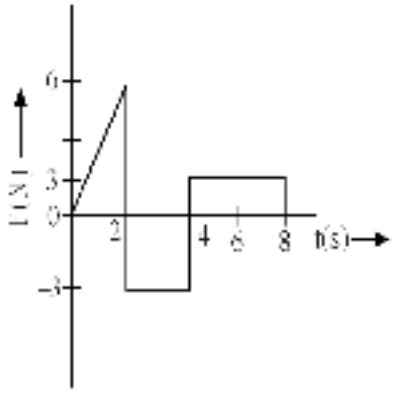
(1) 400 N

(2) 200 N

(3) 300 N

(4) 600 N

41) The force 'F acting on a particle of mass m' is indicated by the force-time graph shown below. The change in momentum of the particle over the time interval from zero to 8 s is



(1) 24 Ns

(2) 20 Ns

(3) 12 Ns

(4) 6 Ns

42) A 1 kg particle strikes a wall with velocity 1 m/s at an angle of 30° with the normal to the wall and reflects at the same angle. If it remain in contact with wall for 0.1 s, then the force is-

(1) 0

(2) $10\sqrt{3}$ N

(3) $30\sqrt{3}$ N

(4) $40\sqrt{3}$ N

43) A ball of mass 0.15 kg is dropped from a height 10 m, strikes the ground and rebounds to the same height. The magnitude of impulse imparted to the ball is ($g = 10\ m/s^2$) nearly :

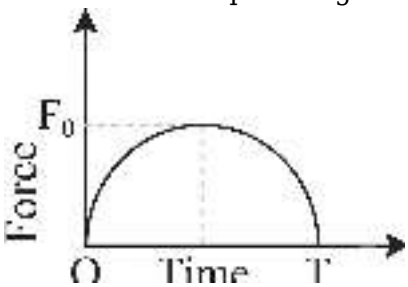
(1) 0 kg m/s

(2) 4.2 kg m/s

(3) 2.1 kg m/s

(4) 1.4 kg m/s

44) A particle of mass m, initially at rest, is acted upon by a variable force F for a brief interval of timeT. It begins to move with a velocity u after the force stops acting. F is shown in the graph as af function of time. The curve is a semicircle.



(1) $u = \frac{\pi F_0^2}{2m}$

(2) $u = \frac{\pi T^2}{8m}$

(3) $u = \frac{\pi F_0 T}{4m}$

(4) $u = \frac{F_0 T}{2m}$

45) A machine gun fires a bullet of mass 60 gm with velocity 1000 ms^{-1}. If average force acting on gun is 200N then find the number of bullet fired per minute

(1) 200

(2) 240

(3) 280

(4) 320

CHEMISTRY

1) What is incorrect about  and  :-

(1) Both are the members of same homologous series

(2) Both are unsaturated

(3) Both have similar general formula

(4) Both have similar molecular formula

2) Identify heterocyclic compound from the following:-

(i) Cyclohexanol (ii) Aniline

(iii)  (iv) 

(1) (i) & (ii)

(2) (i) & (iv)

(3) (iii) & (iv)

(4) (ii) & (iv)

3) Which of the following is incorrect IUPAC name?

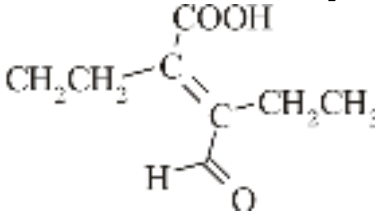
(1) 1–ethyl–2, 2–dimethyl cyclohexane

(2) 2–methyl pentanoic acid

(3) 2–ethyl–3–methyl pentanoic acid

(4) 2–formyl butanoic acid

4) Number of carbon in parent carbon chain in the following compound



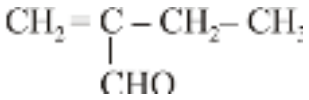
(1) 4

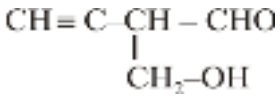
(2) 5

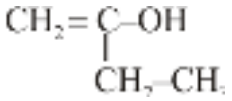
(3) 6

(4) 3

5) Which of the following contain minimum number of C–atom in PCC :

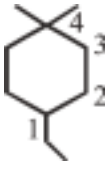
(1) 

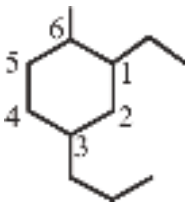
(2) 

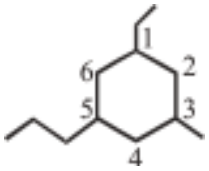
(3) 

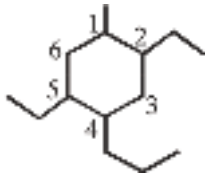
(4) 

6) In which of the following compound IUPAC numbering is correct ?

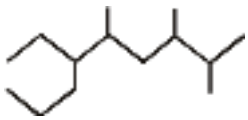
(1) 

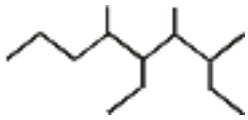
(2) 

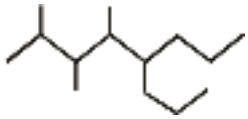
(3) 

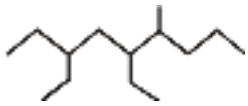
(4) 

7) The correct structure of 6-ethyl-2,3,5-trimethyl nonane is :-

(1) 

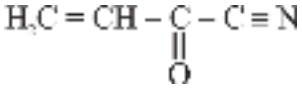
(2) 

(3) 

(4) 

8)

The no. of σ and π bonds present in given compound is.



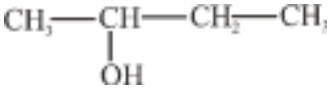
(1) 8 σ and 4 π

(2) 10 σ and 5 π

(3) 8 σ and 3 π

(4) 9 σ and 4 π

9) Which of the following is pair of Homologous series ?

(1) $CH_3-CH_2-CH_2-CH_2-OH$, 

(2) $CH_3-CH_2-CH_2-CH_2-NH_2$,
$CH_3-CH_2-NH-CH_3$

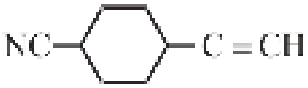
(3) CH_3-SH , CH_3-CH_2-OH

(4) CH_3-O-CH_3 , $CH_3-CH_2-O-CH_3$

10) How many carbon are present in 3rd member of ester:-

(1) 2
(2) 3
(3) 4
(4) 5

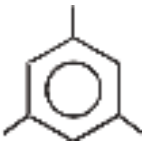
11) Which of the following compound has sp hybridised C and olefinic bond ?

(1) 
(2) 
(3) 
(4) 

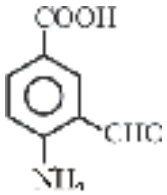
12) The correct IUPAC name of tertiary Butyl amine is :-

(1) Propanamine
(2) 2-methyl propan-2-amine
(3) 2,2-Dimethyl ethanamine
(4) Tertiary Butanamine

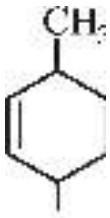
13) Which of the following have only 2°C atoms ?

(a)  (b) 
(c)  (d) 

(1) a, b, c, d
(2) b, c
(3) a, b
(4) a, b, d

14)  has correct IUPAC name :

(1) 1-Amino-2-Formyl benzoic acid
(2) 4-Amino-3-Formyl benzoic acid
(3) 2-Amino-5-Carboxy benzaldehyde
(4) 2-Formyl-4-Carboxy aniline

15)  has correct IUPAC name

(1) 1-Methyl Cyclohexane-3-ol

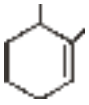
(2) 3- Methyl Cyclo hex-2-en-1-ol

(3) 4-Methyl Cyclohex-2-en-1-ol

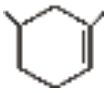
(4) 1-Hydroxy-4-methyl Cyclohex-2-ene

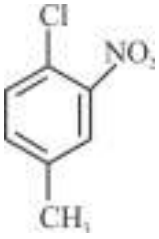
16) Which of the following name will be incorrect :-

(1)  3,6– Dimethyl cyclohexene

(2)  1,6– Dimethyl cyclohexene

(3)  6,6–Dimethyl cyclohexene

(4)  1,5–Dimethyl cyclohexene

17) Correct IUPAC name for compound is :- 

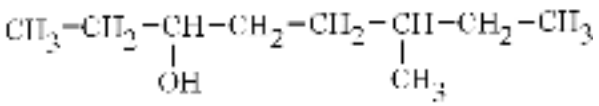
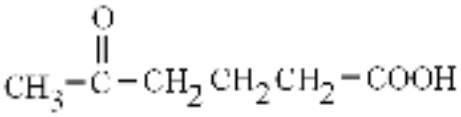
(1) 1-Chloro-2-nitro-4-methylbenzene

(2) 2-Chloro-1-nitro-5-methylbenzene

(3) 1-Chloro-4-methyl-2-nitrobenzene

(4) m-Nitro-p-chlorotoluene

18) In which of the following given IUPAC name is wrong :

(1)		6-methyl octan-3-ol
(2)		5-oxohexanoic acid

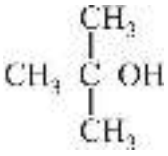
(3)	$HC\equiv C–CH=CH–CH=CH_2$	Hexa-3, 5-diene-1-yne
(4)		Cyclohex-2-en-1-ol

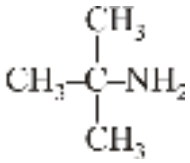
(1) 1

(2) 2

(3) 3

(4) 4

19) **Statement-I :-**  is tertiary alcohol.

Statement-II :-  is tertiary amine.

(1) Both Statement-I and Statement-II are incorrect.

(2) Statement-I is correct but Statement-II is incorrect.

(3) Statement-I is incorrect but Statement-II is correct.

(4) Both Statement-I and Statement-II are correct.

20) Match the column X and Y :-

	X (Organic compounds)		**Y (General formula)**
(P)	Saturated aldehydes and ketones	(I)	C_nH_{2n-2}
(Q)	Saturated aliphatic amides	(II)	$C_nH_{2n}O$
(R)	Alkanols	(III)	$C_nH_{2n+1}\cdot NO$
(S)	Alkadienes	(IV)	$C_nH_{2n+2}\cdot O$

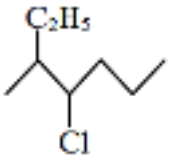
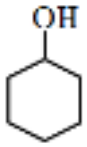
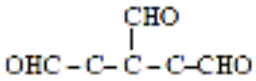
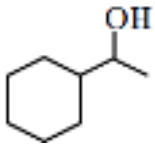
(1) (P)-(II), (Q)-(III), (R)-(IV), (S)-(I)

(2) (P)-(III), (Q)-(II), (R)-(IV), (S)-(I)

(3) (P)-(II), (Q)-(I), (R)-(III), (S)-(IV)

(4) (P)-(I), (Q)-(II), (R)-(III), (S)-(IV)

21)

	Column-I Compound		**Column-II** no. of C in parent carbon chain
A		P	6
B		Q	7
C		R	3
D		S	2

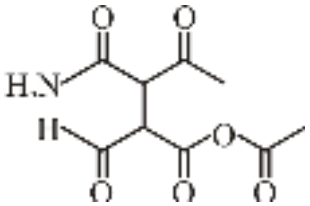
	A	B	C	D
(1)	P	Q	R	S
(2)	S	R	Q	P
(3)	Q	P	R	S
(4)	Q	R	S	P

(1) 1

(2) 2

(3) 3

(4) 4



22) Calculate total number of functional group in following structure?

(1) 4

(2) 5

(3) 6

(4) 7

23) Which is incorrect match

	Common Name	**IUPAC Name**

(1)	Malonic acid	Propanedioic acid
(2)	Isobutyric acid	2-Methyl propanoic acid
(3)	Phenyl acetic acid	2-Phenyl ethanoic acid
(4)	Mesityl oxide	3-Methyl pent-3-en-2-one

(1) 1

(2) 2

(3) 3

(4) 4

24)

The mechanism of the reaction,
$2NO(g) + 2H_2(g) \rightarrow N_2(g) + 2H_2O(g)$ is :

Step 1 : $2NO(g) + H_2(g) \xrightarrow{slow} N_2 + H_2O_2$

Step 2 : $H_2O_2 + H_2 \xrightarrow{fast} 2H_2O$

Then the correct statement is-

(1) $Rate = k[NO]^2[H_2]^2$

(2) $Rate = k[H_2O_2][H_2]$

(3) On doubling the concentration of H_2, keeping the concentration, of 'NO' constant, the rate will become double

(4) None of these

25)

The time for half life period of a certain reaction A → Products is 1 hour. When the initial concentration of the reactant 'A', is 2.0 mol L^{-1}, how much time does it take for its concentration to come from 0.50 to 0.25 mol L^{-1} if it is a zero order reaction ?

(1) 0.25h

(2) 1h

(3) 4h

(4) 0.5h

26) In a reaction $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$, the rate of appearance of NH_3 is 2.5×10^{-4} $molL^{-1}$ sec^{-1}. The rate of reaction & rate of disappearance of H_2 will be (In mol L^{-1} sec^{-1})

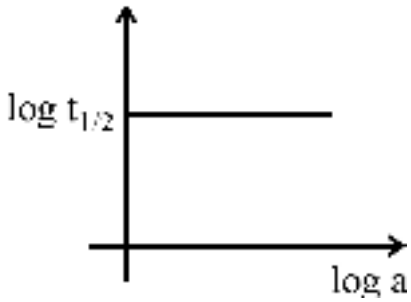
(1) 3.75×10^{-4}, 1.25×10^{-4}

(2) 1.25×10^{-4}, 2.5×10^{-4}

(3) 1.25×10^{-4}, 3.75×10^{-4}

(4) 5.0×10^{-4}, 3.75×10^{-4}

27) A graph between log $t_{1/2}$ and log a, 'a' being the initial concentration of reactant A is given, for



reaction : A → product. If 75% of A is decomposed in 100 min. How long will it take for 90% of the compound to decompose.
[Given : log 2 = 0.3]

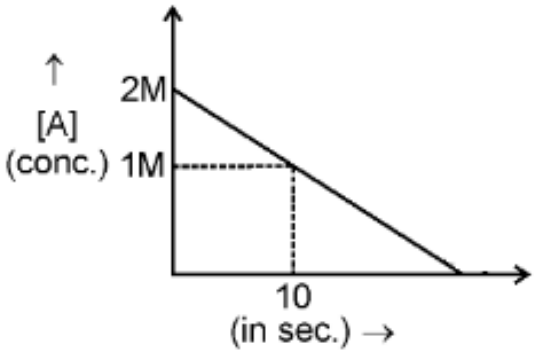
(1) 176.66 min.

(2) 156.66 min.

(3) 166.66 min.

(4) 150 min.

28) For a zero order reaction, $A \xrightarrow{k} P$
Variation of concentration of A with time is given as:



Then calculate time to reduce concentration of A from 2 M to 0.5 M

(1) 10 sec

(2) 20 sec

(3) 15 sec

(4) 25 sec

29) In the following reaction : xA → yB

$$\log_{10}\left[-\frac{d[A]}{dt}\right] = \log_{10}\left[\frac{d[B]}{dt}\right] + \log_{10} 2$$

where –ve sign indicates rate of disappearance of the reactant. Thus, x : y is

(1) 1 : 2

(2) 2 : 1

(3) 3 : 1

(4) 3 : 10

30) Proposed mechanism for decomposition of O_3 is given as :
Net reaction : $2O_3 \rightarrow 3O_2$

$O_3 \underset{K_{-1}}{\overset{K_1}{\rightleftharpoons}} O_2 + O$ (fast equilibrium)

$O + O_3 \xrightarrow{K_2} 2O_2$ (slow)
Some informations are :
(P) Overall order of the reaction is 3/2
(Q) Overall order of the reaction is 1
(R) Increasing O_2 concentration decreases overall rate of reaction.
(S) Decreasing O_3 concentration increases overall rate of reaction.

(1) P and R
(2) Q and R
(3) Q and S
(4) P and S

31) The units for rate constant of first order reaction is–

(1) s^{-1}
(2) Mol $L^{-1}s^{-1}$
(3) Mol s^{-1}
(4) L Mol^{-1} s^{-1}

32) The reaction, X + 2Y + Z → N occurs by the following mechanism

(i)	X + Y ⇌ M	very rapid equilibrium
(ii)	M + Z → P	slow
(iii)	P + Y → N	very fast

What is the rate law for this reaction
(1) Rate = k[Z]
(2) Rate = k[X] $[Y]^2$ [Z]
(3) Rate = [N]
(4) Rate = k[X] [Y] [Z]

33) Given below are two statements
Statement-I : Number of reacting species taking part in an elementary reaction which must collide simultaneously in order to bring about a chemical reaction is called molecularity of a reaction.
Statement-II : Order of reaction is applicable to elementary reaction only, and for complex reaction order has no meaning.
In light of above statements, choose the correct answer from the options given below :

(1) Both statement I and statement II are true.
(2) Both statement I and statement II are false.
(3) Statement I is true but statement II is false.
(4) Statement I is false but statement II is true.

34) For the reaction 2A + B → C, following results are obtained-

$[A]_0$	$[B]_0$	Initial Rate of reaction (in M sec^{-1})
0.1 M	0.1 M	6×10^{-4}
0.4 M	0.1 M	12×10^{-4}
0.4 M	0.2 M	48×10^{-4}

then correct rate equation will be :-

(1) $r = k[A]^2 [B]^2$

(2) $r = k[A]^{1/2} [B]^2$

(3) $r = k[A]^1 [B]^2$

(4) $r = k[A]^2 [B]^1$

35) For a 1st order reaction, ratio of time required for 99.9% completion to time required for 50% completion is :-

(1) 5

(2) 2

(3) 10

(4) 9

36) For a gas phase reaction
$R(g) \rightarrow 2P(g) + Q(g)$
if initial pressure of R(g) is 1 atm and after 40 sec, total pressure of gas mixture becomes 1.8 atm, then rate of appearance of P in this interval is :

(1) 1×10^{-2} atm/sec

(2) 2×10^{-2} atm/sec

(3) 4×10^{-2} atm/sec

(4) 8×10^{-2} atm/sec

37) The rate equation of a reaction is as follows :
$r = k [A]^{0.5}[B] [C]^{0.5}$
Which statement about the above equation is wrong ?

(1) Order with respect to each A and C is 0.5

(2) Unit of specific reaction rate is $mol^{-1}L\ s^{-1}$

(3) Order with respect to B is one

(4) Total order of the reaction is 2.5

38) Match the column I with column II :

	Column I		**Column II**
(P)	Second order reaction	(1)	Molecularity > 1

(Q)	Pseudo first order reaction	(2)	$[A]_t = [A]_0\, e^{-kt}$
(R)	Zero order reaction	(3)	$t_{100\%} = \frac{[A]_0}{k}$
(S)	First order reaction	(4)	$t_{\frac{1}{2}} \propto \frac{1}{[A]_0}$

Correct code is :

(1) P-3, Q-4, R-1, S-2

(2) P-4, Q-1, R-3, S-2

(3) P-4, Q-3, R-1, S-2

(4) P-2, Q-1, R-4, S-3

39) Find order of reaction from following data :

Initial conc. (M)	1.6	0.8	0.4
Half life	2 min	4 min	8 min

(1) 1

(2) 2

(3) 3

(4) zero

40) The rate constant for a first order reaction is $4.606 \times 10^{-3}\ S^{-1}$. The time required to reduce 2.0 g of the reactant to 0.2 g is :

(1) 1000 s

(2) 100 s

(3) 2000 s

(4) 500 s

41) From different sets of data of $t_{1/2}$ at different initial concentrations say 'a' for a given reaction, [$t_{1/2} \propto a$] is found. The order of reaction is :–

(1) 0

(2) 1

(3) 2

(4) 3

42) Elementry reaction between A and B is second order reaction. Which of following relation is correct-

(1) $r = k[A]^2[B]^0$

(2) $r = k[A]^0[B]^2$

(3) $r = k[A]^{3/2}[B]^{1/2}$

(4) $r = k[A]\,[B]$

43) Calculate the half life of the first order reaction
$C_2H_4O(g) \rightarrow CH_4(g) + CO(g)$
If initial pressure of $C_2H_4O(g)$ is 80mm of Hg and the total pressure at the end of 20 minutes is 120mm of Hg.

(1) 20 min

(2) 10 min

(3) 40 min

(4) 80 min

44)

For a zero order reaction
$A \longrightarrow B$; $K = 0.025\ Ms^{-1}$
If 2M of A is taken initially the concentration of A after 40 sec is -

(1) 0.5M

(2) 1 M

(3) 0.25 M

(4) 2M

45) The following data were obtained during a first order reaction

$$A_{(g)} \longrightarrow B_{(g)} + C_{(g)}$$

Calculate rate constant for this 1st order reaction :-

Exp. No.	Time (in sec.)	Total Pressure (in atm)
1.	0	0.4 atm
2.	50 sec	0.6 atm

(1) $1.38 \times 10^{-3}\ sec^{-1}$

(2) $1.38 \times 10^{-2}\ sec^{-1}$

(3) $1.38 \times 10^{-4}\ sec^{-1}$

(4) $1.28 \times 10^{-5}\ sec^{-1}$

BIOLOGY

1) A bacteriophage known as φ ×174 has ....A.... nucleotides, where as Bacteriophage lambda has ....B.... base pairs .
Choose the correct word for ‘A’ and ‘B’ blanks

	'A'	'B'
1	48502	3.3×10^6
2	53806	485002
3	48502	5386

4	5386	48502

(1) 1

(2) 2

(3) 3

(4) 4

2) Match List - I with List - II :

	List-I		**List-II**
(a)	Pitch of DNA	(i)	O.34 nm
(b)	Distance between two base pair	(ii)	3.4 nm
(c)	Diameter of DNA	(iii)	36°
(d)	Each step of ascent	(iv)	2 nm

(1) (a) – (iii), (b) – (ii), (c) – (iv), (d) – (i)

(2) (a) – (iii), (b) – (iv), (c) – (ii), (d) – (i)

(3) (a) – (ii), (b) – (i), (c) – (iv), (d) – (iii)

(4) (a) – (iv), (b) – (ii), (c) – (i), (d) – (iii)

3) Length of a double strand DNA is 17000 angestrom. Find out number of base pairs and N glycosidic bonds respectively ?

(1) 5000, 10000

(2) 10000, 20000

(3) 10000, 5000

(4) 20000, 10000

4) DNA is made up of polynucleotide chain. A nucleotide of DNA has which of the following three components :-

(1) A nitrogenous base, a ribose sugar and a phosphate group

(2) A nitrogenous base, a deoxyribose sugar and a sulphate group

(3) A nitrogenous base, a deoxyribose sugar and a phosphate group

(4) A nitrogenous base, a ribose sugar and a sulphate group

5) Which of the following statement is not correct ?

(1) Cytosine is common for both DNA and RNA

(2) Thymine is common for both DNA and RNA

(3) Adenine is common for both DNA and RNA

(4) Guanine is common for both DNA and RNA

6) In eukaryotic cell chromatin fibers that are further coiled and condensed at metaphase stage of cell division to form :-

(1) Nucleoid

(2) Nucleolus

(3) Euchromatin

(4) Chromosomes

7) Given below are two statements:
Statement I: A typical nucleosome contains 200 bp of DNA helix
Statement II: In a typical nucleus, some region of chromatin are loosely packed are referred to as euchromatin.
In the light of above Statement choose the correct answer from the options given below:

(1) Statement I is correct but Statement II is incorrect

(2) Both Statement I and Statement II are incorrect

(3) Statement I is incorrect but Statement II is correct

(4) Both Statement I and Statement II are correct

8) Match List - I with List - II :

	List-I		**List-II**
(a)	Histone	(i)	Loosely packed chromatin
(b)	Nucleosomes	(ii)	DNA wrapped around histone octamer
(c)	Euchromatin	(iii)	Tightly packed chromatin
(d)	Heterochromatin	(iv)	Positive charged basic protein

(1) (a) – (iii), (b) – (ii), (c) – (iv), (d) – (i)

(2) (a) – (ii), (b) – (iv), (c) – (iii), (d) – (i)

(3) (a) – (i), (b) – (iii), (c) – (iv), (d) – (ii)

(4) (a) – (iv), (b) – (ii), (c) – (i), (d) – (iii)

9) Who proposed that for a double stranded DNA, the ratios between Adenine and Thymine and Guanine and Cytosine are constant and equals one:-

(1) Erwin Chargaff

(2) Wilkins

(3) Franklin

(4) Griffith

10) Which of the following technique was used to elucidate the double helix model of DNA

(1) Electromagnetic radiation

(2) UV spectroscopy

(3) X-ray diffraction

(4) Gamma radiation

11) Given below are two statements:

Statement I: In dsDNA the backbone is constituted by sugar-phosphate, and the bases project inside.
Statement II: The two chains of dsDNA are coiled only in a left-handed fashion.
In the light of above Statement choose the correct answer from the options given below:

(1) Statement I is correct but Statement II is incorrect

(2) Statement I is incorrect but Statement II is correct

(3) Both Statement I and Statement II are correct

(4) Both Statement I and Statement II are incorrect

12) Given below are two statements:
Statement I: one strand of DNA which has a free phosphate moiety, referred to as 3’-end of the chain.
Statement II: one Strand of DNA which has a free OH group referred to as 5' -end of the the chain.
In the light of above Statement choose the correct answer from the options given below:

(1) Statement I is correct but Statement II is incorrect

(2) Statement I is incorrect but Statement II is correct

(3) Both Statement I and Statement II are correct

(4) Both Statement I and Statement II are incorrect

13) Which of the following is not a correct statement

(1) Adenylic acid is a Nitrogen Base

(2) Deoxyguanosine is a nucleoside

(3) Cytosine is a nitrogen base

(4) Uridine is a nucleoside

14) Which of the following is not a correct statement regarding to nucleic acid ?

(1) Nitrogen bases are heterocyclic compound

(2) In RNA ribose sugar is present

(3) Adenine and Guanine are substituted purines

(4) For nucleic acid the building block is amino acid

15) Given below are two statements one is labelled as Assertion(A) and the other is labelled as Reason(R).
Assertion(A): If the sequence of bases in one strand of DNA is known then the sequence in other strand can be predicted
Reason(R): Both strands of DNA are complementary to each other
In the light of above statements, choose the most appropriate answer from the options given below :

(1) (A) is not correct but (R)is correct

(2) Both (A) and (R)are correct and (R)is the correct explanation of (A)

(3) Both (A) and (R)are correct and (R)is not the correct explanation of (A)

(4) (A) is correct but (R)is not correct

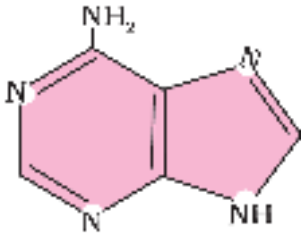
16) Read the following statements and choose the incorrect set of statements:
(a) In prokaryotes, such as, E. coli, though they do not have a defined nucleus, and the DNA is scattered throughout the cell.
(b) In prokaryotic cell DNA in nucleoid is organised in large loops held by histone proteins.
(c) Histones are rich in the basic amino acid residues lysine and arginine
(d) In prokaryotic cell nucleosomes in chromatin are seen as 'beads-on-string' structure when viewed under electron microscope

(1) (a) and (b) only
(2) (a),(b) and (d) only
(3) (a) and (d) only
(4) (b) and (d) only

17) One of the strands of double stranded DNA has base composition as follows : 15% A, 10% T, 45% G and 30% C. What will be the percentage of these bases in the complementary strand?

(1) 15% A, 10% T, 45% G, 30% C
(2) 10% A, 15% T, 45% G, 30% C
(3) 15% A, 10% T, 30% G, 30% C
(4) 10% A, 15% T, 30% G, 45% C

18) Identify the given diagram and select the correct option



(1) Adenine :- found in nucleic acid
(2) Guanine :- found only protein
(3) Adenylic acid :- found only in DNA
(4) Adenosine :- found only in RNA

19) Which of the following bonds are not found in double stranded DNA
(A) N- Glycosidic bond
(B) Disulphide bond
(C) Phosphodiester bond
(D) Peptide Bond
(E) Hydogen Bond

(1) Only (B) ,(C) and (E)
(2) Only (B), (D) and (E)
(3) Only (B) and (D)
(4) Only (A) ,(B) and (D)

20) Which one is **wrong** statement about genetic material?

(1) It should be able to express itself in the form of Mendelian characters

(2) DNA chemically is less reactive and structurally more stable than RNA

(3) RNA being unstable, mutant at a faster rate.

(4) Only DNA is able to mutate not RNA

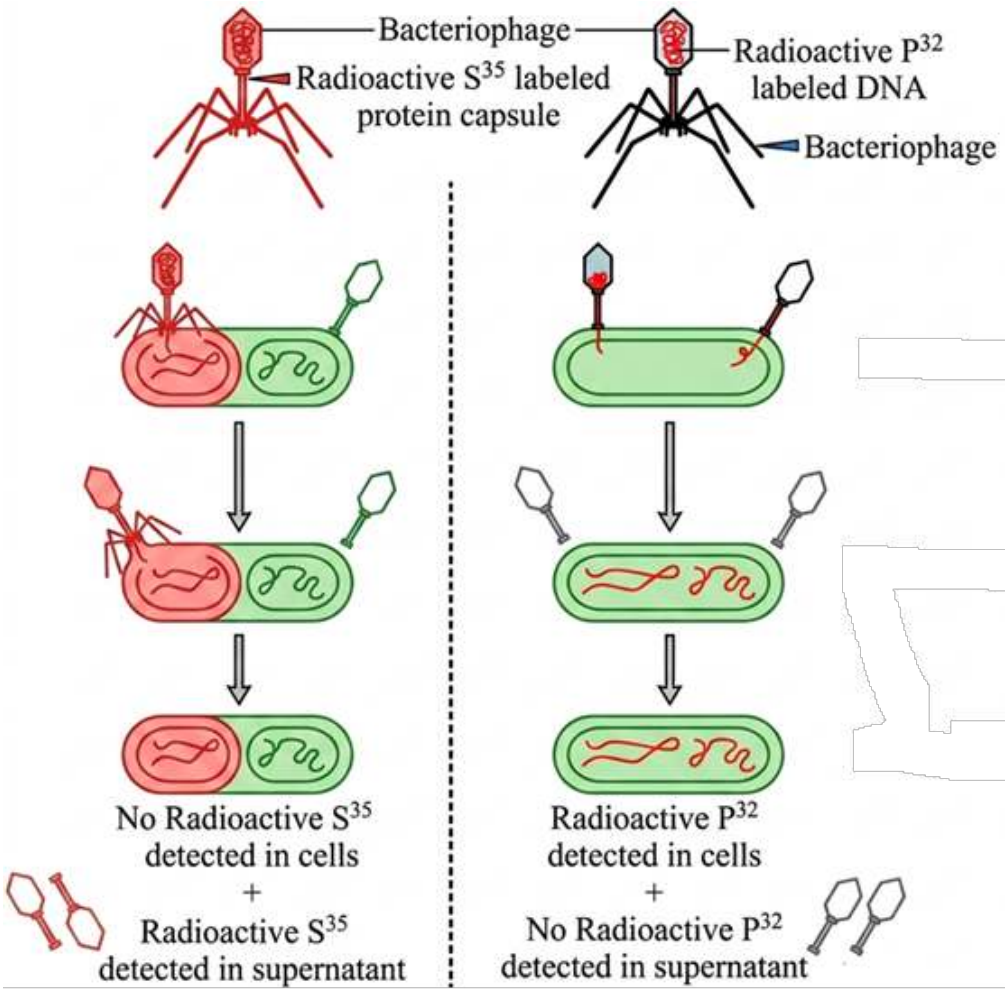
21) There is a definite region in E.coli DNA where the replication originates. Such regions are termed as

(1) origin of replication

(2) Promotor

(3) Terminator

(4) TATA box



22)
This experiment is conducted by :

(1) Frederick Griffith

(2) Hershey and Chase

(3) Avery, Mcleod, McCarty

(4) Watson and Crick

23) A heavy DNA molecule is continuously allowed to replicate in N^{14} medium, then what is the percentage of heavy DNA in 4th generation :-

(1) 25%

(2) 0%

(3) 12.5%

(4) 100%

24) Match the **Column-I** and **Column-II**

Column-I		**Column-II**	
(a)	Structural RNA	(i)	m-RNA
(b)	Messanger RNA	(ii)	TMV
(c)	Adapter RNA	(iii)	r-RNA
(d)	genomic RNA	(iv)	t-RNA

Select the correct option:-

(1) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

(2) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

(3) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

(4) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)

25) Which is not true for RNA :-

(1) RNA is a poly deoxyribonucleotide

(2) In RNA, every nucleotide residue has an additional –OH group present at 2' position in the ribose

(3) RNA is labile and easily degradable

(4) RNA being unstable, so mutate at a faster rate

26) Match the column-I with column-II :-

	Column-I		**Column-II**
(A)	Adapter molecular	(i)	Griffith
(B)	Semiconservative mode of DNA replication	(ii)	Watson & Crick
(C)	Transformation experiment	(iii)	Meselson & Stahl
(D)	Double helix model of DNA	(iv)	Crick

(1) A-iv, B-iii, C-i, D-ii

(2) A-i, B-iv, C-ii, D-iii

(3) A-ii, B-i, C-iii, D-i

(4) A-iv, B-i, C-iii, D-ii

27) Find the **correct** statement with respect to nucleosome :-

(1) Nucleosomes are seen as string on beads structure in chromatin when viewed under light microscope

(2) The core of histone molecules is rich in acidic amino acid residues which carry positive charges in their side chains

(3) A typical nucleosome contains 400 bp of DNA helix

(4) The negatively charged DNA is wrapped around the positively charged histone octamer

28) Consider the following four statements and select the incorrect statement.

(1) Two nucleotides are linked through 3'-5' phosphodiester linkage to form a dinucleotide

(2) A nitrogenous base is linked to the pentose sugar through a ester linkage to form a nucleotide

(3) The plane of one base pair stacks over the other in double helix. This in addition to H-bonds confers stability of the helical structure

(4) The nitrogenous bases linked to sugar moiety project from the backbone

29) When streptococcus pneumoniae bacteria are grown on a culture plate, some produce smooth shiny colonies (S). This is because the S strain bacteria have a :

(1) Histone coat

(2) Mucous coat

(3) Protenious coat

(4) Monosaccharide coat

30) In a typical nucleus , some region of chromatin are loosely packed are referred to as euchromatin. It is said to be in this chromatin.

(1) Replication does not occur

(2) Transcription does not occur

(3) Transcription occur

(4) B-DNA is not present

31) If the length of DNA is 1.36 mm, how many number of nitrogenous bases are presently?

(1) 3.3×10^9

(2) 4.6×10^6

(3) 48502

(4) 8×10^6

32) Which of the following scientist were worked to determine the biochemical nature of transforming principle in Griffith experiments ?

(1) Hershey and Chase

(2) Watson and Crick

(3) Avery, Mac Leod and Mc Carty

(4) Meselson and Stahl

33) Read the following statements and choose the correct set of statements:

(a) The DNA-dependent DNA polymerases catalyse polymerisation only in one direction, that is 5' to 3'.
(b) the replication occur within a small opening of the DNA helix, referred to as replication fork
(c) the average rate of polymerisation by DNA polymerase has to be approximately 2000 bp per second
(d) In E.coli the process of replication completes within 60 minutes
Choose the correct answer from the options given below:

(1) (a), (b), (c) and (d) only

(2) (a), (c) and (d) only

(3) (a) and (d) only

(4) (a), (b) and (c) only

34) In Hershey and Chase experiment which of the following step was not performed ?

(1) Infection

(2) Blending

(3) X-Ray differaction

(4) Centrifugation

35) Which of the following statement is incorrect ?

(1) DNA polymerase uses a DNA template to Catalyse the polymerisation of deoxynucleotides.

(2) DNA template with polarity 3' → 5', the replication is discontinuous.

(3) A failure of cell division after DNA replication lead to polyploidy.

(4) In Hershey and Chase experiment virus grown in the presence of radioactive phosphorous contains radioactive DNA

36) *E.Coli* DNA has only____I____ base pairs and the process of replication complete with in ____II____ minutes.
Choose the correct data for I and II blank.

	I	II
1	6.6×10^9	38
2	48 502	18
3	4.6×10^6	18
4	3.3×10^9	38

(1) 1

(2) 2

(3) 3

(4) 4

37) By using which of the following Taylor experimentally proved that the DNA in chromosomes replicate semiconservatively ?

(1) X - ray diffraction data

(2) Isotop of Nitrogen

(3) Radioactive Sulfur

(4) Radioactive Thymidine

38) Which of the following enzyme facilitates opening of the DNA helix during the process of replication?

(1) DNA Ligase

(2) RNA Polymerase

(3) DNA Polymerase

(4) DNA helicase

39) The diagram shows an important concept in the genetic implication of DNA. Fill in the blanks A to B:

DNA $\xrightarrow{A}$ RNA $\xrightarrow{B}$ Protein

(1) A-translation B-extension

(2) A-transcription B-translation

(3) A-transcription B-replication

(4) A-translation B-transcription

40) In which of the following organism genetic material is RNA :

(1) Lambda bacteriophage

(2) T2 bacteriophage

(3) E.coli

(4) QB bacteriophage

41) Given below are two statements :
Statement-I :- Griffith concluded that the R strain bacteria had somehow been transformed by the heat-killed S strain bacteria.
Statement-II :- In Hershey and Chase experiment bacteria which was infected with viruses that bacteria had radioactive DNA were radioactive, indicating that DNA was the material that passed from the virus to the bacteria.
In the light of above statement choose the correct answer from the option given below :

(1) **Statement I** is correct but **Statement II** is incorrect

(2) Both **Statement I** and **Statement II** are incorrect

(3) **Statement I** is incorrect but **Statement II** is correct

(4) Both **Statement I** and **Statement II** are correct

42) Presence of which of the following nitrogen base in DNA enhance stability over RNA ?

(1) Thymine

(2) Uracil

(3) Cytosine

(4) Adenine

43) If DNA contained sulphur instead of phosphorus and proteins contained phosphorus instead of sulfur, what would have been the outcome of Hershey and chase experiment ?

(1) No radioactive sulfur in bacterial cells

(2) Both radioactive sulfur and phosphorus in bacterial cells

(3) Radioactive sulfur in bacterial cells

(4) Radioactive phosphorus in bacterial cells

44) In Meselson and Stahl experiment after 40 minutes growth of E.coli cell hybrid and light DNA are separated by which of the following techniques :

(1) CsCl density gradient centrifugation

(2) MgCl density gradient centrifugation

(3) X-ray differection pattern

(4) Normal centrifugation

45) Which of the following statement is incorrect with respect to DNA replication ?

(1) Replication is very fast process

(2) Replication is very expensive process

(3) Any mistake during replication would result into mutation

(4) During replication ribonucleoside monophosphate serve dual purpose

46) Which of the following testicular cells provide nutrition to the germ cells ?

(1) Sertoli

(2) Leydig cells

(3) Spermatogonia

(4) Seminal vesicles

47) Which of the following statement is incorrect about the male reproductive system ?

(1) The testis are situated outside the abdominal cavity within a pouch called scrotum.

(2) The scrotum helps in maintaining the high temprature of the testes.

(3) The width of each testis is about 2 to 3 cm.

(4) Each testicular lobule contains one to three coiled seminiferous tubules.

48) Given below are two statements.
Statement-I :- Each seminiferous tubule is lined on its side by two types of cell called male germ cells and sertoli cells.
Statement-II :- The male germ cells undergo meiotic divisions finally leading to sperm formation.

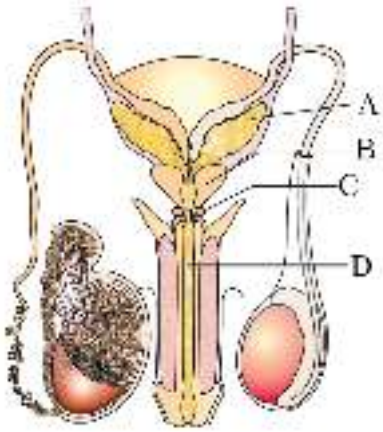
(1) Statement-I and statement-II both are correct.

(2) Statement-I and statement-II both are incorrect.

(3) Statement-I is correct but statement-II is incorrect.

(4) Statement-I is incorrect but statement-II is correct.

49) Find out the correct answer about the part labelled as A, B, C and D from the figure given below :-



	Part-A	Part-B	Part-C	Part-D
(1)	Seminal vesicles	Prostate gland	Bulbourethral glands	Urethra
(2)	Urethra	Seminal vesicles	Prostate gland	Bulbourethral glands
(3)	Bulbourethral glands	Urethra	Seminal vesicles	Prostate gland
(4)	Prostate gland	Bulbourethral glands	Urethra	Seminal vesicles

(1) 1

(2) 2

(3) 3

(4) 4

50) Epididymis is located on :-

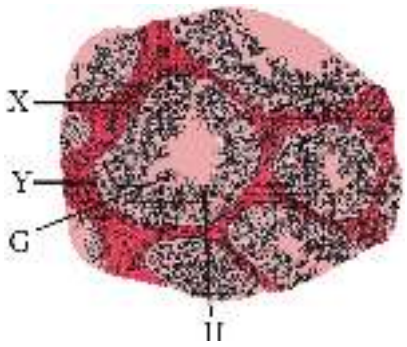
(1) Posterior surface of single testis

(2) Anterior surface of single testis

(3) Posterior surface of each testis

(4) Anterior surface of each testis

51) In the given diagram identify the cell and choose correct statement :-



(1) X is diploid and provide nutrition to germ cells
(2) Y is haploid and forms spermatids
(3) G is haploid and secretes ABP
(4) H is diploid and provide nutrition to germ cells

52) Choose the incorrect regarding testis :-

(1) 250 testicular lobules in each testis
(2) 4–5 highly coiled seminiferous tubules in each lobule
(3) Low temperature than normal internal body temperature
(4) Covered by a dense covering

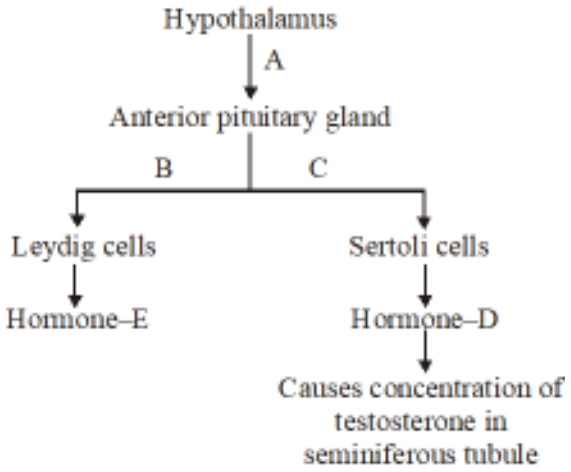
53) Primary oocytes arrest in which phase ?

(1) Prophase–I of meiotic division
(2) Prophase–I of mitotic division
(3) Prophase–I of equational division
(4) Prophase–I of meiotic–II division

54) No. of polar bodies formed in human female before the age of puberty :-

(1) Two
(2) One
(3) Three
(4) Zero

55) Observe the following flow chart and choose right statement :-



(1) B cause release of E and E inhibits secretion of D (normally)

(2) C causes release of D and D concentrates E inside the seminiferous tubule

(3) A is GnRH, C is ICSH and B is FSH

(4) A is GnRH, C is LH and B is FSH

56) Choose the incorrect :-

(1) Spermatogonia $\xrightarrow[\text{Growth}]{\text{Mitosis}}$ primary spermatocytes

(2) Secondary spermatocytes $\xrightarrow[\text{division}]{2^{nd}\text{ mitosis}}$ spermatids

(3) Primary spermatocytes $\xrightarrow[\text{division}]{\text{Reductional}}$ secondary spermatocytes

(4) Spermatids $\xrightarrow[\text{division}]{\text{Reductional}}$ spermatozoa

57) Layer, which line uterine cavity ?

(1) Perimetrium

(2) Myometrium

(3) Thin & membranous

(4) Endometrium

58) Choose the correct statement regarding acrosome

(1) Cap-like structure, present on posterior end of sperm

(2) Present on anterior end of sperm and cup-like structure.

(3) Filled with enzymes that help fertilization of sperm

(4) Cap like structure, present on anterior end of sperm

59) What is the correct sequence of sperm formation?

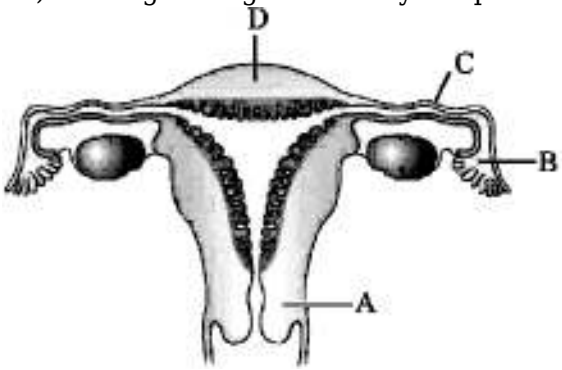
(1) Spermatid, spermatocyte, spermatogonia, spermatozoa

(2) Spermatogonia, spermatocyte, spermatozoa, spermatid

(3) Spermatogonia, spermatozoa,spermatocyte, spermatid

(4) Spermatogonia, spermatocyte, spermatid, spermatozoa

60) In the given figure identify the parts labelled as A, B, C and D and select correct option :-



	A	**B**	**C**	**D**
(1)	Cervix	Infundibulum	Isthmus	Fundus
(2)	Ovary	Ampulla	Isthmus	Fimbriae
(3)	Ovary	Infundibulum	Ampulla	Fundus
(4)	Cervix	Isthmus	Ampulla	Fimbriae

(1) 1

(2) 2

(3) 3

(4) 4

61) The feature of some structures of male reproductive system is given below. Identify the structure on the basis of the characteristics which surrounds the primary sex organ of male reproductive system.

(1) Its enlarged end is called glans penis.

(2) It travels through the penis and carry semen as well as urine.

(3) It is responsible for maintaining the low temperature by about 2-2.5° C from normal body temperature to mature sperm.

(4) Stores sperms prior to ejaculation.

62) **Assertion :** Surgical removal of bulbourethral glands may affects the process of copulation.
Reason : The secretions of bulbourethral glands helps in the lubrication of the penis.

(1) Assertion and reason both are true but reason is not correct explanation of assertion

(2) Assertion and reason both are true and the reason is correct explanation of assertion

(3) Assertion is true but reason is wrong

(4) Assertion and reason both are wrong

63) **Assertion :** The male sex accessory ducts include rete testis, vasa efferentia, epididymis and vas deferens
Reason : The male accessory glands include paired seminal vesicles, a prostate and paired bulbourethral glands.

(1) Assertion and reason both are true and the reason is correct explanation of assertion

(2) Assertion and reason both are true but reason is not correct explanation of assertion

(3) Assertion is true but reason is wrong

(4) Assertion and reason both are wrong

64) Read column I & column II and identify their correct match :-

Column-I			**Column-II**
A.	spermatogonia	I.	Provide nutrition to germ cells
B.	seminal plasma	II.	male germ cells
C.	leydig cells	III.	present outside of seminiferous tubule.
D.	sertoli cells	IV.	secretion of seminal vesicle
		V.	secretion of seminal vesicles, prostate & bulbourethral gland
		VI.	spermatid
		VII.	contains fructose, calcium & certain enzyme

(1) A-II, B-IV,VII, C-III, VI D-III, I

(2) A-II, B-V,VII, C-III, D- I

(3) A-II, B-VII, C-I, VI D-III

(4) A-II, B-IV, C-VI, D-VII

65) Choose the correct pathway of milk ejection from inside to outside in the mammary gland :-

(1) Mammary duct → Mammary tubule → Ampulla → Lactiferous duct

(2) Alveolus → Mammary tubule → Mammary duct → Ampulla

(3) Alveolus → Mammary duct → Lactiferous duct → Ampulla

(4) Ampulla → Alveolus → Mammary duct → Lactiferous duct

66) Match the following columns and choose the correct option :-

	Column I		**Column-II**
(A)	Insemination	(i)	Differentiation of spermatids into spermatozoa
(B)	Spermiation	(ii)	Transfer of sperms into the female genital tract
(C)	Capacitation	(iii)	Changes in the sperm before fertilisation
(D)	Spermiogenesis	(iv)	release of sperms from seminiferous tubules

(1) A-i, B-i, C-iii, D-iv

(2) A-iii, B-iv, C-ii, D-i

(3) A-iii, B-i, C-ii, D-iv

(4) A-ii, B-iv, C-iii, D-i

67) Match column-I with column-II.

	Column-I		**Column-II**
A.	Labia minora	i.	Cushior of fatty tissue
B.	Mons pubis	ii.	Upper junction of labia minora
C.	Clitoris	iii.	Paired fold of tissue
D.	Hymen	iv.	Membrane of mucosal tissue

(1) A-i, B-ii, C-iii, D-iv

(2) A-iii, B-i, C-ii, D-iv

(3) A-ii, B-iii, C-i, D-iv

(4) A-iii, B-iv, C-i, D-ii

68) **Assertion (A) :** The presence or absence of the hymen is a reliable indicator of a woman's virginity.
Reason (R) : The hymen can be torn due to various reasons other than sexual intercourse, such as physical activity or the use of tampons.

(1) 'Both A and R are true and R is the correct explanation of A'

(2) 'Both A and R are true but R is not the correct explanation of A'

(3) 'A is true but R is false'

(4) 'A is false but R is true'

69) Read the following statements :-
(a) Seminal plasma along with sperm constitute the semen.
(b) The human male ejaculate about 2-3 million sperms during a coitus.
(c) The middle piece of sperm possesses numerous mitochondria
(d) The sperm head contains an round haploid nucleus
Choose the correct options :-.

(1) a,c

(2) a,d

(3) b,c

(4) a,b

70) **Assertion :** Seminiferous tubules is a structural and functional unit of ovary
Reason : Each testis has about 250 compartments called testicular lobules and each lobule contains one to three highly coiled seminiferous tubules and it from sperms

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is false statement but Reason is true

(4) Both Assertion and Reason are false statements

71) **Assertion :** Ovarian follicles in different stages of development are embedded in the stroma
Reason : The female external genitalia includes mons pubis, labia majora, labia minora, hymen and clitoris

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false.

72) Sperm entry into the ovum induces the completion of the meiotic division of the :-

(1) Primary oocyte
(2) Secondary oocyte
(3) Ootid
(4) Mature ovum

73) To produce 100 million spermatozoa, how many secondary spermatocytes are required?

(1) 50 million
(2) 25 million
(3) 100 million
(4) 200 million

74) Read Column-I & Column-II and identify their correct match :-

	Column-I		**Column-II**
(A)	At the age of puberty	(i)	Formation of secondary oocyte
(B)	Prior to ovulation	(ii)	Formation of primary oocyte
(C)	During embryonic period	(iii)	Formation of second polar body
(D)	At the time of fertilisation	(iv)	Formation of first polar body
		(v)	Spematogenesis starts
		(vi)	Oogenesis starts
		(vii)	Completion of Meiosis-I of primary oocyte
		(viii)	Completion of Meiosis-II of secondary oocyte.
		(ix)	Formation of haploid ovum (ootid)

(1) A-v,vi, B-i,iii, C-ii,iv,viii, D-vii, ix

(2) A-v, B-i, iv, vii, C-ii, vi, D-viii, ix

(3) A-v, B-i, vii, ix, C-ii, iv, D-iii, vi, vii

(4) A-v, vi, B-ii, iv, vii, C-i, iii, D-viii, ix

75) Read following terminology of column-I & II and identify their correct match :-

	Column-I		**Column-II**
(A)	Spermatogonia	(i)	Haploid
(B)	Primary spermatocyte	(ii)	44 = XY
(C)	Secondary spermatocyte	(iii)	Diploid
(D)	Sperm	(iv)	Haploid
		(v)	Formed due to meiosis I
		(vi)	Formed due to growth
		(vii)	Acrosome

(1) A-ii, B-iv, v, C-iii, vi, D-i, vii

(2) A-ii, B-iii, vi, C-iv,v, D-i, vii

(3) A-i,vii B-iii, vi, C-iv, v, D-ii

(4) A-i, B-ii, iii, C-iv, v, D-vi, vii

76) Match the list - I with list-II.

	List-I		**List-II**
(A)	Uterus	(i)	Finger like projections
(B)	Clitoris	(ii)	Inverted pear shaped
(C)	Fimbriae	(iii)	Highly coiled
(D)	Seminiferous tubules	(iv)	Tiny finger like structure

(1) A (ii), B (i), C (iv), D (iii)

(2) A (ii), B (iii), C (iv), D (i)

(3) A (ii), B (i), C (iii), D (iv)

(4) A (ii), B (iv), C (i), D (iii)

77) Read the following terminology of column-I, II, & III and identify their correct match :-

	Column-I		**Column-II**		**Column-III**
(A)	Scrotum	(P)	Extra abdominal pouch	(i)	Sperm production
(B)	Seminiferous tubules	(Q)	Accessory gland	(ii)	Provide nutrition to sperms
(C)	Sertoii cells	(R)	Highly coiled	(iii)	Paired structure

(D)	Seminal vesicle	(S)	Inside the seminiferous tubules.	(iv)	Maintains the low temperature of tests.

(1) A-P-iv, B-R-i, C-S-ii, D-Q-iii

(2) A-P-iv, B-R-i, C-S-iii, D-Q-ii

(3) A-P-i, B-R-iv, C-S-iii, D-Q-ii

(4) A-R-i, B-Q-iii, C-P-ii, D-S-iv

78) During implantation, the embryo becomes embedded in which layer of uterus ?

(1) Perimetrium

(2) Myometrium

(3) Endometrium

(4) Epimetrium

79) Which of the following is a content of spermatic cord?

(1) Dartos muscle

(2) Cremaster muscle

(3) Vas deferens

(4) Vasa efferentia

80) Liberation of sperms from sertoli cells is called :-

(1) Semination

(2) Spermiation

(3) Ejaculation

(4) Insemination

81) Which accessory sex gland in human being is unpaired ?

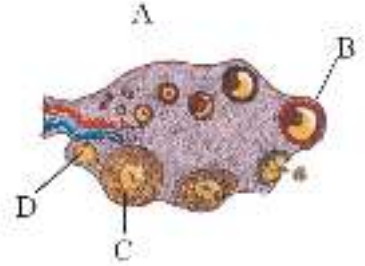
(1) Bartholin's gland

(2) Seminal vesicle

(3) Cowper gland

(4) Prostate gland

82) The figure below shows ovarian changes during a mestruation cycle (A, B, C, D). Select the correct structure which secretes progesterone for maintenance of pregnancy.



(1) B

(2) D

(3) A

(4) C

83)

Proximal centriole donated by sperm during fertilization is used for

(1) Completion of meiosis

(2) First division of zygote

(3) Fast block polyseprmy

(4) None of these

84)

Arrange following events of female reproductive cycle (human) in sequence :-
(I) Secretion of FSH
(II) Growth of corpus luteum
(III) Growth of follicles
(IV) Ovulation

(V) LH Surge

(1) III → I → IV → II → V

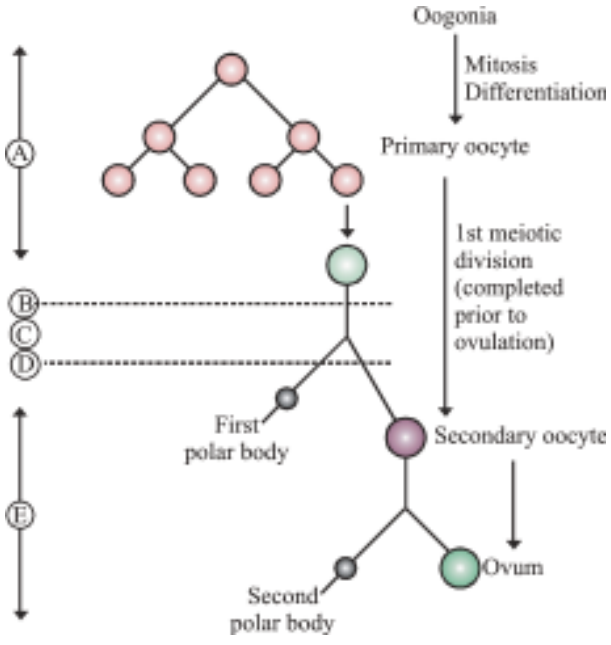
(2) I → III → V → IV → II

(3) I → IV → III → V → II

(4) II → I → III → IV → V

85)

The following refers to schematic representation of oogenesis.



Identify A to E correctly.

(1) (A) Fetal life, (B) Birth, (C) Puberty, (D) Adult reproductive life, (E) Child hood

(2) (A) Fetal life, (B) Birth, (C) Child hood, (D) Puberty, (E) Adult reproductive life
(3) (A) Adult reproductive life, (B) Birth, (C) Puberty, (D) Child hood, (E) Fetal life
(4) (A) Birth, (B) Child hood, (C) Fetal life, (D) Puberty, (E) Adult reproductive life

86) Identify the structure shown below and its location in the human, and select the right option.



	Structure	**Site of occurrence**
(1)	Testis	Abdominal cavity
(2)	Penis	Abdominal cavity
(3)	Seminiferous tubules	Testis
(4)	Prostate	Below the urinary bladder

(1) 1
(2) 2
(3) 3
(4) 4

87) When released from ovary, human egg contains:-

(1) X/Y chromosome
(2) 2 X chromosome
(3) 1 X chromosome
(4) XY chromosome

88) Which of the following statement is incorrect?

(1) The male germ cells undergo meiotic divisions finally leading to sperm formation
(2) Sperm and ovum formation ceases around the age of fifty years
(3) In fact, the presence or absence of hymen is not a reliable indicator of virginity
(4) The functions of male sex accessory ducts and glands are maintained by the testicular hormones

89)

If in a sperm, proximal centriole becomes

non-functional, which of the following process will be affected?
(1) Fertilization
(2) Spermiogenesis
(3) Movement of sperm

(4) Cleavage

90) Which of the following is not considered as secondary sexual character in females ?

(1) Breasts
(2) Broad pelvis
(3) Pubic hairs
(4) Vagina

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	4	4	1	3	4	4	3	3	3	4	1	4	1	4	3	3	4	4	3	1
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	3	3	1	1	1	3	2	2	2	3	1	2	2	2	2	1	4	1	4	4
Q.	41	42	43	44	45															
A.	3	2	2	3	1															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	4	3	1	2	1	3	1	1	4	3	2	2	3	2	3	3	3	3	2	1
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	3	1	4	3	1	3	3	3	2	2	1	4	3	2	3	2	4	2	2	4
Q.	86	87	88	89	90															
A.	1	4	1	2	2															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	4	3	1	3	2	4	4	4	1	3	1	4	1	4	2	2	4	1	3	4
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	1	2	2	3	1	1	4	2	2	3	4	3	4	3	2	3	4	4	2	4
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	1	3	1	4	1	2	1	1	3	4	2	1	4	2	4	4	4	4	1
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	3	2	2	2	2	4	2	4	1	3	2	2	1	2	2	4	1	3	3	2
Q.	171	172	173	174	175	176	177	178	179	180										
A.	4	4	2	2	2	3	3	2	4	4										

SOLUTIONS

PHYSICS

1) Slope of line, m = –tan 30° = $-\frac{1}{\sqrt{3}}$
Intercept on y–axis, c = 4
Equation of line, y = mx + c
$\Rightarrow y = -\frac{1}{\sqrt{3}}x + 4$

2)

Slope intercept form
$\frac{x}{6} + \frac{y}{8} = 1$
$\frac{4x + 3y}{24} = 1$
3y + 4x = 24

3) As tan θ is decreasing slope is decreasing

4)

Distance formula

5) **Explain the question :** In this question we have to find the value of given integral.
Concept : This question is based on definite integration.
Solution : $\int_{x_1}^{x_2} a.x^n.dx = a\left(\frac{x^{n+1}}{n+1}\right)_{x_1}^{x_2}$ where 'a' is some constant
$\int_{x_1}^{x_2} \cos x.dx = (\sin x)_{x_1}^{x_2}$
$\int_0^1 9x^8.dx + \int_0^{\pi/2} \cos x.dx = \left(\frac{9x^{8+1}}{8+1}\right)_0^1 + (\sin x)_0^{\pi/2}$
$= \left(x^9\right)_0^1 + (\sin x)_0^{\pi/2}$
$= \left[(1)^9 - (0)^9\right] + \left[\sin\left(\frac{\pi}{2}\right) - \sin(0)\right]$
= 1 + 1 = 2
Final answer : 4

6)

$y = e^{\sin x}$

$\frac{dy}{dx} = e^{\sin x}, x \cos x$

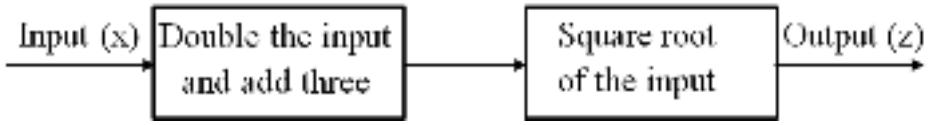
7) $\int_{-\pi/2}^{\pi/2} \cos x$

$= [\sin x]_{-\pi/2}^{\pi/2}$

$= \sin\left(\frac{\pi}{2}\right) - \sin\left(\frac{-\pi}{2}\right)$

$= y + 1 = 2$

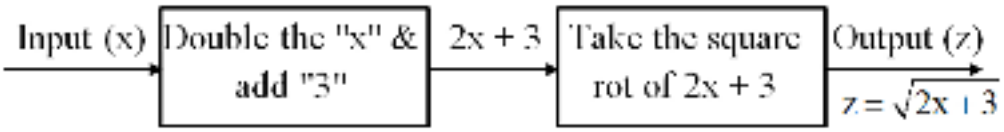
8) **Explain the question :** A function machine is shown as below -



What will be the relation between output (z) and input (x).

Concept: This question is based on functions & relations

Solution :



Final answer : 3

9) **Question Explanation :** Determine the value of $\sqrt{1.04}$ = ?

Concept : Approximation of Binomial Expansion

Formula :

If ⟹ x <<<<< 1

Then ⟹ $(1 + x)^n \approx (1 + nx)$

Solution :

$\sqrt{1.04} = (1.04)^{1/2}$

$= (1 + 0.04)^{1/2}$

$\approx (1 + \frac{1}{2} \times 0.04)$ {∵ 0.04 <<<<< 1}

$\approx (1 + 0.02)$

≈ 1.02

Final Answer : Option (3)

10) **Explain the question :** Evaluate ∴ sin(23°) = ?

Concept : This question is based on the sum & difference formula of sine

Formula : sin(A – B) = sin(A).cos(B) – cos(A).sin(B)

Solution : sin(23°) = sin(60° – 37°)

= sin(60°).cos(37°) – cos(60°).sin(37°)

$= \frac{\sqrt{3}}{2} \times \frac{4}{5} - \frac{1}{2} \times \frac{3}{5}$

$= \frac{2\sqrt{3}}{5} - \frac{3}{10}$

$= \frac{4\sqrt{3}-3}{10}$

Final answer : 4

11) $\int \left(\sqrt{t} + \frac{1}{\sqrt{t}}\right)^2 dt$

$= \int \left(t + \frac{1}{t} + 2\right) dt$

$\frac{t^2}{2} + \log_e t + 2t + c$

12) $\frac{dx}{dt} = 2t$

13)

a = 7
d = 3
n = 7

$S_n = \frac{7}{2}[7 + 25]$

$S_n = \frac{7}{2}[32]$

$S_n = 112$

14) $\frac{dr}{dt} = 0.1$ $A = \pi r^2$

$\frac{dA}{dt} = 2\pi r \cdot \frac{dr}{dt} = \frac{2\pi \times 5}{11} \times 0.1$

$= 0.29 \text{ cm}^2/\text{sec}$

15) $v = 6t^2 - 6t^3 \Rightarrow \frac{dv}{dt} = 12t - 18t^2$

For max. and min., $\frac{dv}{dt} = 0$

$\Rightarrow 12t - 18t^2 = 0$

$6t(2 - 3t) = 0$

$t = 0, t = 2/3$

$\frac{d^2v}{dt^2} = 12 - 3t$

at t = 0, $\frac{d^2v}{dt^2} = 12$

and $t = \frac{2}{3}, \frac{d^2v}{dt^2} = -12$

thus v is max at t = $\frac{2}{3}$ and min. at t = 0

16) $\int ydx = \text{Area}$

$\frac{1}{2} \times 25 \times 20$

= 250

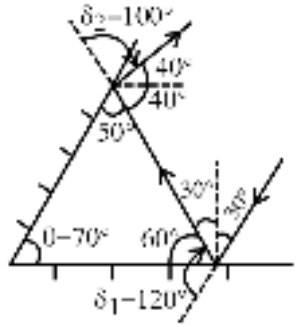
17)

$y = \sec^2 x - \tan^2 x = 1$

$\frac{dy}{dx} = 0$

18) $\delta = 180° - 60° = 120°$

19) **Explain the Question :** Find angle of deviation.
Concept : This question is based on deviation by plane mirror.
Solution :



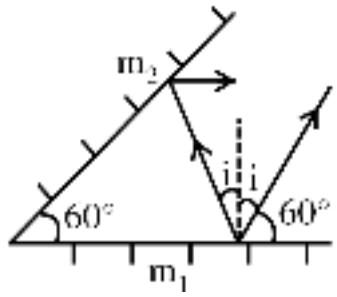
$\delta_1 = 180° - 60° = 120°$ CW
$\delta_2 = 180° - 80° = 100$ CW
$\delta_{net} = \delta_1 + \delta_2$
$\delta_{net} = 120° + 100 = 220°$ CW
Final answer : Option (3)

20) **Explain the question :** The question asks how much the reflected ray turns when both the incident ray and the plane mirror are rotated by specific angles.
Concept : This question is based on effect of mirror rotation.
Solution : When a plane mirror is rotated by an angle θ, the reflected ray turns by 2θ. Additionally, if the incident ray is turned by an angle ϕ, the total angle turned by the reflected rays is:
$= 20°\downarrow + 5°\uparrow = 15°\downarrow$
or
Total angle = 2θ + ϕ.
θ = 10°, ϕ = –5°. So, total angle = 2 × 10° – 5° = 15°.

Final answer : 1

21)

Explain the Question : As shown in figure find angle i.
Concept : This question is based on reflection by plane mirror (deviation).
Solution :

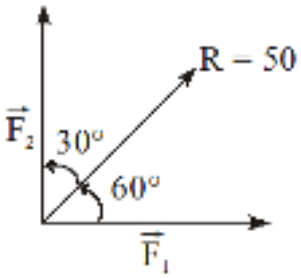


incident ray is parallel to the mirror M_2
$90° = i + 60°$
$i = 90° - 60° = 30°$
Final answer : 3

22) Eye is most sensitive for yellow colour

23) $A_x = A_y = A_z$
$A^2 = 3A_x^2$
$A_x = \frac{A}{\sqrt{3}} = \frac{30}{\sqrt{3}} = 10\sqrt{3}$ units

24) $\bar{P} + \bar{Q} = 11\hat{i} + 9\hat{j}$
$\bar{P} = 7\hat{i} + 6\hat{j}$
$\bar{Q} = 4\hat{i} + 3\hat{j}$
$|\bar{Q}| = 5$



25)
$R \cos 60° = F_1$
$F_1 = 50 \times \frac{1}{2} = 25$
$R \cos 30° = F_2$
$F_2 = 50 \times \frac{\sqrt{3}}{2} = 25\sqrt{3}$

26)

Sum of 3 forces can be zero only if (A–B) ≤ C ≤ A + B.

27) Projection of velocity along $\vec{a}$

$$\left(\frac{\vec{v}\cdot\vec{a}}{a}\right)\left(\frac{\vec{a}}{a}\right)$$

$$\Rightarrow \left(\frac{6+2-2}{\sqrt{3}}\right)\left(\frac{i+\hat{j}+\hat{k}}{\sqrt{3}}\right) = \frac{6}{3}\left(i+j+\hat{k}\right)$$

$$= 2\hat{i}+2\hat{j}+2\hat{k}$$

28) $\vec{R} = \vec{A} - 2\vec{B} + 3\vec{C}$

$$= \left(2\hat{i}+\hat{j}\right) - 2\left(3\hat{j}-\hat{k}\right) + 3\left(6\hat{i}-2\hat{k}\right)$$

$$= 2\hat{i}+\hat{j}-6\hat{j}+2\hat{k}+18\hat{i}-6\hat{k}$$

$$= 20\hat{i}-5\hat{j}-4\hat{k}$$

29) $R = 2a\cos\frac{\theta}{2}$ $\quad 1 = 2 \times 1 \cos\frac{\theta}{2}$

$\theta = 120°$

$R^1 = 2a \sin\frac{\theta}{2} = 2 \times 1 \sin 60° = \sqrt{3}$

30) **Explain Question:**
To find angle between $\vec{A}$ and $\vec{B}$

Concept:
Addition and subtraction of vectors

Solution:

$$\sqrt{A^2 + B^2 + 2AB\cos\theta} = \frac{1}{2} \times \sqrt{A^2 + B^2 - AB\cos\theta}$$

$4(A^2 + B^2 + 2AB\cos\theta) = A^2 + B^2 - 2AB\cos\theta$

$3A^2 + 3B^2 + 10AB\cos\theta = 0$

$12B^2 + 3B^2 + 20B^2\cos\theta = 0$

$20B^2\cos\theta = -15B^2$

$$\cos\theta = -\frac{3}{4}$$

$$\theta = \cos^{-1}\left(-\frac{3}{4}\right)$$

Final answer: Option (3)

31) As $\left|\vec{A}+\vec{B}\right| = n\left|\vec{A}-\vec{B}\right|$

then angle between $\vec{A}$ and $\vec{B}$ is

$$\theta = \cos^{-1}\left(\frac{n^2-1}{n^2+1}\right) \Rightarrow \theta = \cos^{-1}\left(\frac{9-1}{9+1}\right) = 37^\circ$$

32) Let magnitude of $\vec{P}$ is x.

then $\vec{P} + \vec{Q} = 2x\hat{j}$

$\vec{Q} = 2x\hat{j} - x\hat{i}$

$\left|\vec{Q}\right| = \sqrt{(2x)^2 + (-x)^2} = 8$

$x = \frac{8}{\sqrt{5}}$

33)

Area of parallgram = $\frac{1}{2}\left|\vec{d}_1 \times \vec{d}_2\right|$

Now,

$$\vec{d}_1 \times \vec{d}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 4 & 1 \\ 1 & 1 & 1 \end{vmatrix}$$

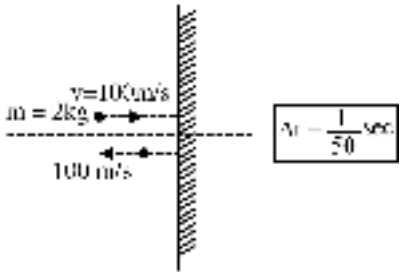
$\vec{d}_1 \times \vec{d}_2 = i(4-1) - j(3-1) + k(3-4)$

$= 3\hat{i} - 2\hat{j} - \hat{k}$

$\left|\vec{d}_1 \times \vec{d}_2\right| = \sqrt{9+4+1} = \sqrt{14}$

So, area = $\frac{\sqrt{14}}{2} = \sqrt{\frac{7}{2}}$

34) **Explain the Question :** Asking about the average force applied on the wall by the body.
Concept : This question is based on striking & rebouncing perpendicularly.
Solution :



$F = \frac{\Delta p}{\Delta t} = \frac{2[100-(-100)]}{\frac{1}{50}}$

$F = 2 \times 10^4$ N

Final Answer: Option (2)

35) **Explain Question:**
Question ask to find gas ejected per second.

Concept:

work with variable force.

Solution:

$$\frac{Vdm}{dt} - mg = 0$$

$$1000\frac{dm}{dt} = mg$$

$$1000\frac{dm}{dt} = 600(10)$$

$$\frac{dm}{dt} = 6kg/sec.$$

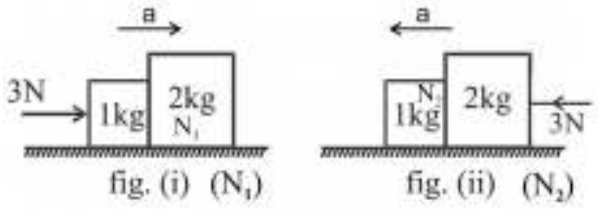
Final Answer: option(2)

36)

Question Explanation: Find the ratio of the normal contact forces $\frac{N_1}{N_2}$ between two blocks of masses 1 kg and 2 kg placed on a smooth surface, when a horizontal force of 3 N is applied to the 1 kg block in fig (i) and to the 2 kg block in fig (ii).

Concept: Newton's Second Law of Motion

Solution: When a force is applied to a system of blocks in contact, they move with a common acceleration, and the internal normal force between them depends on which mass is being pushed.



$$a = \frac{3}{3} = 1m/s^2$$

$N_1 = 2a \qquad N_2 = 1a$

$$\frac{N_1}{N_2} = \frac{2}{1}$$

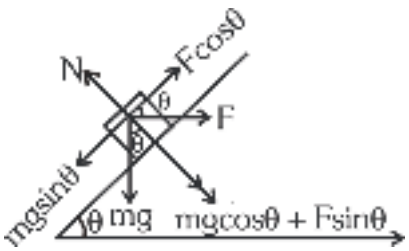
Final answer: Option (1)

37)

Question Explanation : We need to find magnitude of normal reaction acting on the block, shown in the figure.

Concept : Free body diagram (FBD)

Solution :



From the figure, the normal reaction on the block is N = mgcosθ + F sinθ.

Final answer : Option (4)

38) **Question Explanation:** Calculating contact forces between three blocks.
Concept: System acceleration and Free Body Diagrams.
Solution: Total mass = 3 + 2 + 1 = 6kg

Acceleration (a) = $\frac{12N}{6kg}$ = $2m/s^2$

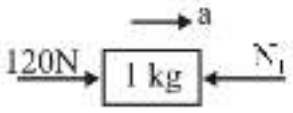
f_2 (force on 3kg block) = m × a = 3 × 2 = 6N

f_1 (force between 2kg and 1kg) pushes (3kg + 2kg)

f_1 = (3 + 2) × a = 5 × 2 = 10N

Final Answer: Option (1)

39)



$$a = \frac{120-50}{(1+4+2)} = \frac{70}{7} = 10m/s^2$$

$(120 - N_1) = (1)(a)$

$N_1 = 120 - 10$

N_1 = 110 newton

40) N = mg + ma = m (g+a) = 40 (10+5) = 600 N

41)

Area of f-t curve = ΔP

$\Delta P = \frac{1}{2} \times 2 \times 6 - 2 \times 3 + 4 \times 3$

= 6 – 6 + 12

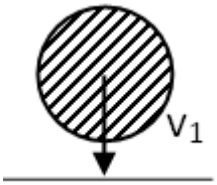
= 12 Ns

42) $F = \frac{\Delta p}{\Delta t} = \frac{2mv\cos 30^\circ}{\Delta t} = \frac{2 \times 1 \times 1 \times \sqrt{3}/2}{0.1} = 10\sqrt{3}N$

43)

Velocity just before striking the ground $v_1 = \sqrt{2gh}$

$v_1 = \sqrt{2(10)(10)} = 10\sqrt{2}$ m/s ; $\vec{v}_1 = -10\sqrt{2}\,\hat{j}$

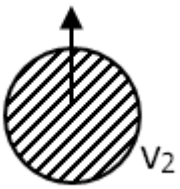


If it reaches the same height, speed remains same after collision only the direction changes.

$v_2 = 10\sqrt{2}$ m/s

$\vec{v_2} = 10\sqrt{2}\,\hat{j}$

$|\text{Impulse}| = m|\Delta\vec{v}|$



$= m|10\sqrt{2}\,j - (-10\sqrt{2}\,j)|$

$= 0.15\left[2(10\sqrt{2})\right] = 3\sqrt{2}$ kg m/s = 4.2 kg m/s

44)

Area of f-t curve = ΔP

$\Rightarrow \frac{1}{2} \times [\pi] \times \frac{1}{2} \times F^0 = m\,[u - 0]$

$= \frac{F_0 T \pi}{4} = mu$

$= \boxed{u = \frac{\pi F_0 T}{4m}}$

45) $F = \frac{\Delta p}{\Delta t} = n\,(mu)$

$200 = n \times \left[\frac{60}{100} \times 1000\right]$

$n = \frac{200}{60}$ Bullet/sec.

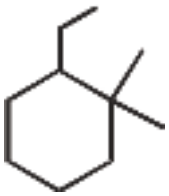
So, number of bullet fired per minute

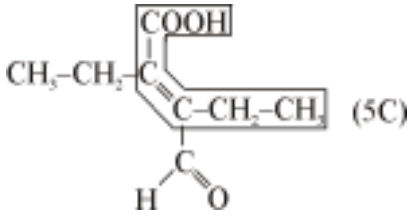
$= \frac{200}{60} \times 60 = 200$

CHEMISTRY

46)  

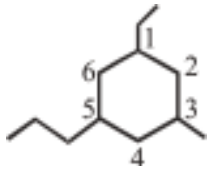
47)  &  are heterocyclic compound.

48)  2–ethyl–1,1–dimethyl cyclohexane

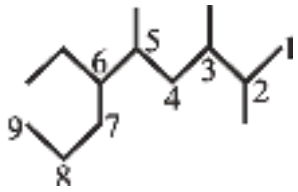
49) 

(5C)

50) 1 = 3C 2. 4C 3. 4C 4.4C

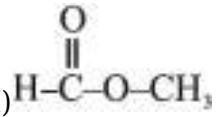
51) 

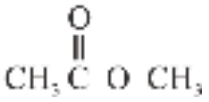
Positions are same follow alphabatical order.

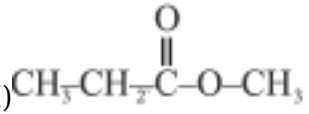
52) 

53) 8σ and 4π bond

54) Members of Homologous series have same functional group and difference of $–CH_2$ or $2(CH_2)$ or $3(CH_2)$ or so on.

55) (I) 

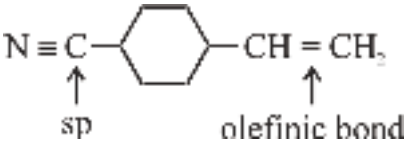
(II) 

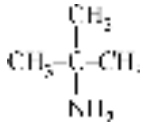
(III) 

(4C)

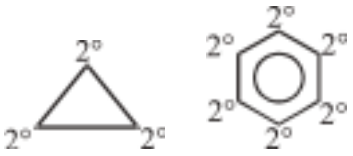
56)

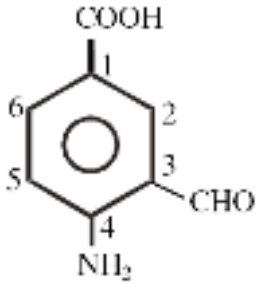
Olefinic bond is C = C



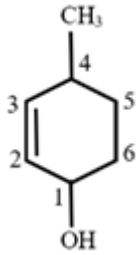
57) 

Correct IUPAC Name is 2-methyl propan-2-amine.

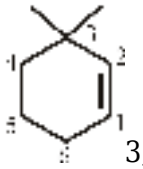
58) 

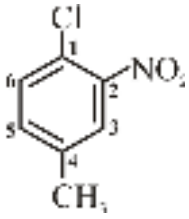
59) 

60)



4-Methyl Cyclohex-2-en-1-ol
Final answer: option (3)

61)  3,3–Dimethyl cyclohexene

62) 

63)

Correct IUPAC name of HC ≡ C – CH = CH – CH = CH_2 is Hexa-1,3-diene-5-yne.

64)

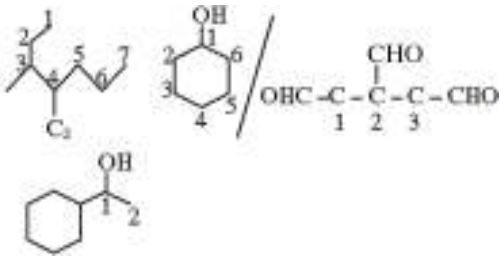
Statement :- I Is correct because degree of alcohol is decided by degree of carbon at which –OH group is attached**.**
So here –OH is directly attached to 3°C Thus it is tertiary alkohol
Statement -II is incorrect because degree of amine is decided by number of carbon which is directly attached to nitrogen, so here one carbon is directly attached to nitrogen Thus it is primary amine.
Final answer : Option (2)

65)

Explain the Question : Match the column X and Y.
Concept : This question is based on
Solution : General formula can be obtained by consider that alkanes have : CnH_{2n+2}.
For 1π bond : –2H
1'N': +1H
1 'O' : no change in 'H'
1 'X': –1 H

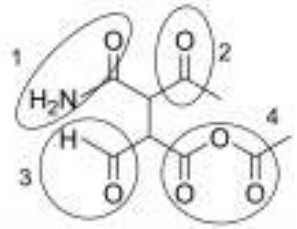
	X (Organic compounds)		**Y (General formula)**
(P)	Saturated aldehydes and ketones	(II)	$C_nH_{2n}O$
(Q)	Saturated aliphatic amides	(III)	$C_nH_{2n+1}\cdot NO$
(R)	Alkanols	(IV)	$C_nH_{2n+2}\cdot O$
(S)	Alkadienes	(I)	C_nH_{2n-2}

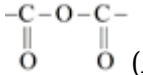
Final answer: 1

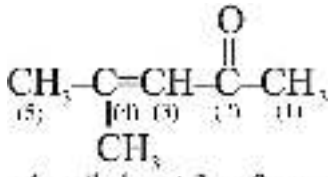
66) 

67)

Question Explanation : Calculate total number of functional groups in given structure.
Concept: IUPAC priority table.

Solution: 

functional groups are
(1) - $CONH_2$ (Amide)
(2) >C=O (Ketone)
(3) -CHO (Aldehyde)
(4)  (Anhydride)
Final answer: option (1)

68) Mesityl oxide 

69)

Slow step is rate determining
$\therefore \text{Rate} = K[NO]^2 [H_2]$

On doubling the concentration of H_2 without changing the concentration of 'NO', the rate will become double

70) **Question Explanation:** Find how much time does it take for concentration to come from 0.50 to 0.25 mol L^{-1}

Concept: This question is based on zero order reaction.

Solution:

$$\frac{t_{0.5_2}}{t_{0.5_1}} = \left[\frac{A_2^0}{A_1^0}\right]$$
$$\frac{t_{0.5_2}}{1} = \left[\frac{0.5}{2}\right]$$
$$t_{0.5_2} = 0.25h$$

Final Answer: Option (1)

71)

According to stoichiometry of reaction :
$$\frac{[\text{rate of disappearance of } N_2]}{1} = \frac{(\text{rate of disappearance of } H_2)}{3} = \frac{(\text{rate of appearance of } NH_3)}{2}$$
= rate of reaction
$\Rightarrow$ rate of reaction
$$= \frac{2.5 \times 10^{-4}}{2} 2 = 1.25 \times 10^{-4} \frac{mol}{\ell - sec}$$
$$\Rightarrow \frac{\text{rate of disappearance of } H_2}{3}$$

$$= 1.25 \times 10^{-4} \frac{\text{mol}}{\ell - \text{sec}}$$

⇒ rate of disappearance

$$= 3.75 \times 10^{-4} \frac{\text{mole}}{\ell - \text{sec}}$$

72) $K = \frac{1}{t} \ln \frac{100}{100 - x}$

$K = \frac{1}{100} \ln 4$ .....(1)

$K = \frac{1}{t} \ln \frac{100}{10}$ .....(2)

From eq. (1) and (2)

$\frac{1}{100} \ln 4 = \frac{1}{t} \ln 10$

$t = \frac{100}{2 \times 0.3}$

73)

For zero order: $t_{3/4} = 1.5\ t_{1/2}$

74) $-\frac{1}{x} \frac{d[A]}{dt} = +\frac{1}{y} \frac{d[B]}{dt}$

75) $K_{eq} = \frac{K_1}{K_{-1}} = \frac{[O_2]\ [O]}{[O_3]}$

$r = K_2\ [O_3]\ [O] = K_2 \cdot [O_3] \cdot \frac{K_1}{K_{-1}} \cdot \frac{[O_3]}{[O_2]}$

$= K \cdot [O_3]^2\ [O_2]^{-1}$

76)

Based on fact

77) Rate = k_1 [M] [Z] .....(1)

from equation (1) $K_{eq} = \frac{[M]}{[x][y]}$.

M = k_{eq} [x] [y] .....(2)

put the value of M from (2) to (1)

Rate = $k_1\ k_{eq}$ [x] [y] [z]

Rate = k [x] [y] [z]

78) **Explain Question:** The question tests understanding of molecularity and order of reaction and asks which statement is true.

Concept: Molecularity, Order of Reaction

Solution: Molecularity: Number of molecules or atoms that must collide in an elementary reaction. Always a whole number.
Order of Reaction: The sum of powers of concentration terms in the rate law. Can be fractional, zero, or whole. Determined experimentally, applicable to all reactions, simple or complex.
Statement I is true: It correctly defines molecularity.
Statement II is false: Order of reaction is not limited to elementary reactions; it applies to complex reactions as well.
Final Answer: Option 3

79)

Question Explanation: Using initial rate data, determine the **order with respect to A and B**, hence the correct rate law.

Concept: Rate law: $r = k[A]^m[B]^n$
Compare experiments by changing **one concentration at a time**.

Solution:

From Exp-1 → Exp-2 (B constant = 0.1 M):
[A] : 0.1 → 0.4 (×4)
Rate : $6\times10^{-4} \rightarrow 12\times10^{-4}$ (×2)
$\Rightarrow 4^m = 2 \Rightarrow m = 1/2$

From Exp-2 → Exp-3 (A constant = 0.4 M):
[B] : 0.1 → 0.2 (×2)
Rate : $12\times10^{-4} \rightarrow 48\times10^{-4}$ (×4)
$\Rightarrow 2^n = 4 \Rightarrow n = 2$

Rate equation: $r = k[A]^{1/2}[B]^2$

Final Answer: Option (2)

80)

Question Explanation : We must find the ratio of times for 99.9% and 50% completion of a first-order reaction.

Concept : This question is based on the first-order integrated rate law:

$$t = \frac{2.303}{k} \log\left(\frac{[A]_0}{[A]_t}\right)$$

Solution :
Given Formula:

$$t = \frac{2.303}{k} \log\left(\frac{[A]_0}{[A]_t}\right)$$

for 99.9% completion: $[A]_t = 0.001 \cdot [A]_0$

$$t_1 = \frac{2.303}{k} \log\left(\frac{1}{0.001}\right)$$

$$t_1 = \frac{2.303}{k} \log(1000)$$

$$t_1 = \frac{2.303}{k} \times 3$$

for 50% completion: $[A]_t = 0.5[A]_0$.

$$t_2 = \frac{2.303}{k} \log(2) = \frac{2.303}{k} \times 0.3$$

Ratio:

$$\frac{t_1}{t_2} = \frac{10}{1} = 10$$

Final Answer : (3)

81)

Question Explanation: Find the rate of appearance of P given the change in total pressure for the reaction R → 2P + Q.

Concept: this question is based on (relation between change in total pressure and rate of reaction).

Solution:

	R(g) →	2P(g) +	Q(g)
t = 0	1 atm	0	0
t = 40	1 – x	2x	x

Given total pressure (P_T) at 40 sec = 1.8 atm

$1 - x + 2x + x = 1.8$

$2x = 0.8$

$x = 0.4$

$$(\text{R.O.A.})_P = \frac{2x}{t} = \frac{2 \times 0.4}{40} \text{atm/sec}$$

$= 2 \times 10^{-2}$ atm/sec

Final Answer : Option (2)

82)

A: Question Explanation

Identify the incorrect statement related to the given rate law:

$r = k [A]^{0.5} [B]^1 [C]^{0.5}$

B: Concept
Order of Reaction and Unit of Rate Constant

C: Reasoning

A. Order w.r.t A = 0.5, C = 0.5 → correct

B. Order w.r.t B = 1 → correct

C. Total order = 0.5 + 1 + 0.5 = 2 → so 2.5 is incorrect

D. Unit of rate = mol L^{-1} s^{-1}
→ For overall order 2,
unit of k = L mol^{-1} s^{-1} or mol^{-1} L s^{-1}, which matches option 2 → correct

D: Final Answer
Total order is 2, not 2.5
Correct option: 4

83)

A: Question Explanation
Match reaction types with their corresponding kinetic features or formulas.

B: Concept
Reaction order, rate law expressions, molecularity, and half-life relationships.

C: Matching

A. **P: Second order reaction** → Half-life $\propto \frac{1}{[A]_0}$ → **(4)**

B. **Q: Pseudo first order reaction** → Molecularity > 1 → **(1)**

C. **R: Zero order reaction** → $t_{100} = \frac{[A]_0}{k}$

D. **S: First order reaction** → $[A]_t = [A]_0 e^{-kt}$

D: Final Answer
Correct matching: P-4, Q-1, R-3, S-2
Correct option: 2

84) $\frac{(t_{1/2})_1}{(t_{1/2})_2} = \left(\frac{a_2}{a_1}\right)^{n-1}$

$\frac{2}{4} = \left(\frac{0.8}{1.6}\right)^{n-1}$

$\frac{1}{2} = \left(\frac{1}{2}\right)^{n-1}$

n = 2

85) First order reaction
K(Rate constant) = $4.606 \times 10^{-3}\ s^{-1}$

$$kt = \ell n\frac{A_0}{A_t}$$

$$4.606 \times 10^{-3} \times t = \ell n\frac{2}{0.2}$$

$$\boxed{t = 500\ sec}$$

86) For zero order reaction,

$$t_{1/2} = \frac{[A_0]}{2k} \text{ or } \frac{t_1}{2} \propto [A_0]$$

So given reaction is zero order reaction

87)

Based on fact

88) $C_2H_4O(g) \rightarrow CH_4(g) + CO(g)$

t = 0	80		
t = 20	80 – P	P	P

80 – P + P + P = 120
P = 40
So $t_{1/2}$ is 20mm.

89) **Question Explanation:** Find concentration of A after 40 sec.

Concept: This question is based on zero order reaction

Solution:

A	→	B
a		0
a – x		x

X = Kt = 0.025 × 40 = 1 M
 a – x = 2 – 1 = 1 M

Final Answer: Option (2)

90)

$A \rightarrow B + C$
n = 2

$$k_1 = \frac{2.303}{t}\log\frac{P_0}{2P_0 - Pt}$$

$$= \frac{2.303}{50}\log\frac{0.4}{2 \times 0.4 - 0.6}$$

$$= \frac{2.303}{50} \log 2$$

$$= \frac{2.303 \times 0.30}{50} = \frac{0.693}{50} = 1.38 \times 10^{-2}\ \text{sec}^{-1}$$

BIOLOGY

91) NCERT, Pg. # 80

92) NCERT-XIIth, Pg. # 96,97,98(E), 89,90,91(H)

93) NCERT, Pg. # 81

94) NCERT, Pg. # 80

95) NCERT, Pg. # 80

96) NCERT-XIIth, Pg. # 99,100(E), 92,93(H)

97) NCERT-XIIth, Pg. # 99, 100(E), 92, 93(H)

98) NCERT-XIIth, Pg. # 99,100(E), 92,93(H)

99) NCERT-XIIth, Pg. # 96,97,98(E), 89,90,91(H)

100) NCERT-XIIth, Pg. # 96,97,98(E), 89,90,91(H)

101) NCERT-XIIth, Pg. # 96,97,98(E), 89,90,91(H)

102) NCERT-XIIth, Pg. # 80, 81(E), 89,90(H)

103) NCERT-XIIth, Pg. # 80, 81(E), 89,90(H)

104) NCERT-XIIth, Pg. # 80, 81(E), 89,90(H)

105) NCERT, Pg. # 81

106) NCERT, Pg. # 83

107) NCERT-XIIth, Pg. # 81

108) NCERT, Pg. # 80

109) NCERT, Pg. # 80, 81

110)

NCERT-(XII), Pg. # 86,87,88

111) NCERT, Pg. # 88,89,90

112) **Question Explanation :** The question asks for the scientists who conducted the experiment illustrated in the provided image.
Concept : Search Of Genetic Material
Solution : The experiment is conducted by Hershey and Chase.
Final Answer : Option (2)

113) NCERT-XII, Pg. # 89

114) NCERT Pg. # 109

115)

NCERT-XII, Pg. # 87

116) NCERT, Pg. # 100, 97, 114

117) NCERT Page-99

118) **Question Explanation:** Bond types in nucleotides
Concept: DNA structure – Bonds & linkages
Solution: In a nucleoside, base links to sugar by **N-glycosidic** bond, not an ester linkage.
Final answer: Option **(2)**

119) NCERT XII Pg. No. # 100

120) NCERT XII Pg. No. # 100

121) NCERT XII Pg. No. # 99

122) NCERT XII Pg. No. # 101

123) NCERT, Pg. # 90

124) NCERT XII Pg. No. # 102

125) NCERT XII Pg. No. # 106,103,101

126) NCERT XII Pg. No. # 106

127) NCERT XII Pg. No. # 106

128) NCERT XII Pg. No. # 110

129) NCERT, Pg. # 82

130) NCERT, Pg # 87

131) NCERT, Pg # 85,86

132) NCERT, Pg # 87

133) NCERT, Pg # 86

134) NCERT, Pg # 88,89,90

135) NCERT, Pg # 88,89,90

136) NCERT-XII, Pg. # 27, Para-2.1

137) NCERT-XII, Pg. # 27, Para-2.1

138) NCERT-XII, Pg. # 27, Para-2.1

139) NCERT-XII, Pg. # 27, Para-2.1, Fig.-2.1(b)

140)

NCERT ; Page No. # 24

141)

NCERT ; Page No. # 24

142)

NCERT ; Page No. # 24

143)

NCERT ; Page No. # 33

144)

NCERT ; Page No. # 33

145)

NCERT ; Page No. # 28

146) NCERT ; Page No. # 33

147)

NCERT ; Page No. # 29

148)

NCERT ; Page No. # 32

149)

NCERT ; Page No. # 33

150)

NCERT ; Page No. # 29

151)

NCERT ; Page No. # 27

152)

NCERT ; Page No. # 27

153)

NCERT ; Page No. # 27

154) NCERT Pg. # 27, 28

155) NCERT Pg. # 31

156) NCERT Pg. # 26,61 (E), 28,34 (H)

157) NCERT Pg#30

158)

Explain question : The question asks for understanding of the hymen and myths about virginity.

Concept : Female external genitalia

Solution : The correct answer is option 4.

A. The assertion is false; the hymen is not a reliable indicator of virginity.

B. The Reason is correct; the hymen can be torn by various activities other than sexual intercourse, such as strenuous exercise, horseback riding, or the use of tampons.

Final answer : option (4)

159) NCERT Pg. # 32(E), 35 (H)

160)

NCERT ; Page No. # 28

161)

NCERT ; Page No. # 29

162) NCERT-XII Pg#36

163) NCERT, Pg. # 31

164) NCERT XII Pg. # 31,32,36

165) NCERT Pg. # 31

166)

Explain Question: Matching the structures of reproductive organs with their correct descriptions.

Concept: Human Reproduction – Reproductive Anatomy

Solution: The correct answer is option 4.

A. **Uterus** – (ii) Inverted pear-shaped structure.

B. **Clitoris** – (iv) Tiny finger-like structure.

C. **Fimbriae** – (i) Finger-like projections at the end of fallopian tubes.

D. **Seminiferous tubules** – (iii) Highly coiled structures in testes.

Final Answer: Option (4)

167) NCERT Pg. # 27,28

168)

Explain Question : To identify the layer in which embryo becomes embedded During implantation

Concept : This question is based on implantation.

Solution : During implantation, the embryo becomes embedded in Endometrium of uterus

Final Answer : option (3)

169)

NCERT ; Page No. # 37

170)

Explain Question : To identify the term used for Liberation of sperms from sertoli cells.

Concept : This question is based on Spermatogenesis

Solution : Liberation of sperms from sertoli cells is called Spermiation.

Final Answer : option (2)

171)

NCERT Pg. # 44

172)

Explain Question : To identify the correct Labelling or structure which secretes progesterone for maintenance of pregnancy, In the given figure showing ovarian changes during a menstrual cycle.

Concept : This question is based on Follicular Developement.

Solution : The structure which secretes progesterone for maintenance of pregnancy is corpus leuteum which is here indicated by "C"

Final Answer : option (4)

173)

Question Explanation : Identify the role of sperm’s proximal centriole during fertilization.

Concept: Fertilization

Solution: The correct answer is Option 2 .

A. The proximal centriole from sperm contributes to formation of the mitotic spindle required for the first cleavage division of the zygote.

Final Answer: option (2)

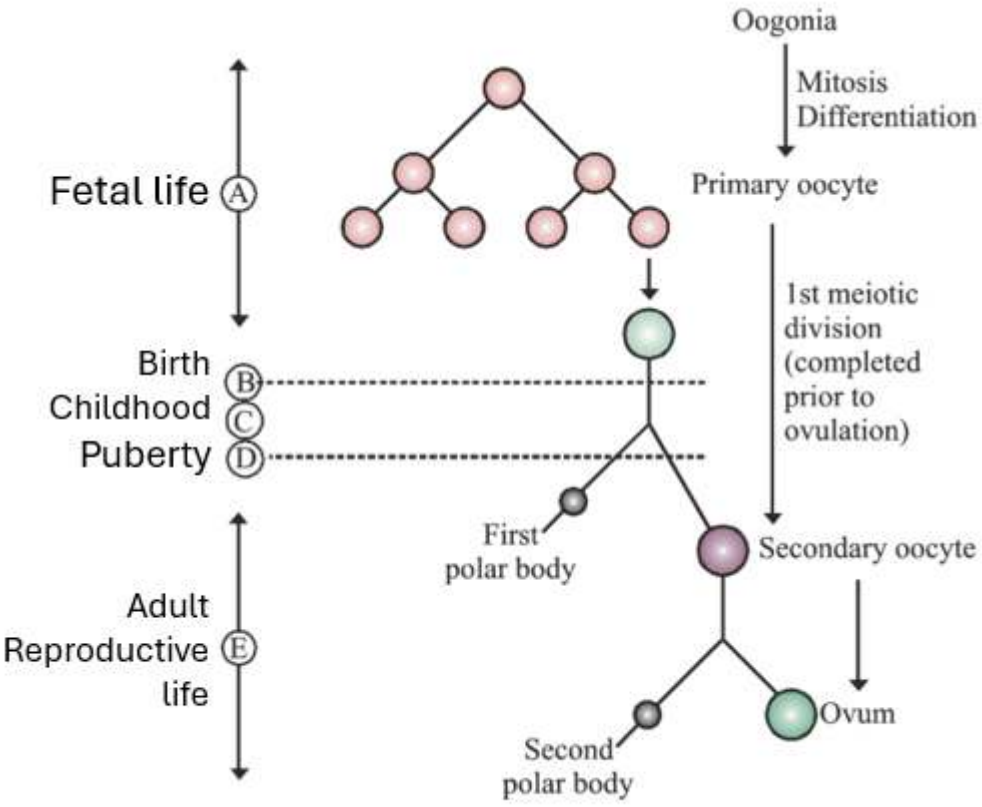
174)

I → III → V → IV → II

175) **Explain Question :** The following refers to schematic representation of oogenesis. Identify A to E correctly.

Concept : This question is based on Oogenesis

Explanation :



Final answer : Option 2

176)

NCERT ; Page No. # 28

177)

Explain Question : to identify the set of chromosome that a human egg contains When released from ovary.

Concept : This question is based on oogenesis.

Solution : When released from ovary, human egg contains 1 X chromosome only

Final Answer : option (3)

178) **Explain Question :** This question is asking about the incorrect statement related to human reproductive biology.
Concept : Reproductive System and Gametogenesis
Explanation : In males, sperm formation continues throughout life after puberty, while in females, ovum formation ceases around the age of fifty. Hence, the statement that both sperm and ovum formation stop at fifty years is incorrect.
Final answer : (2) Sperm and ovum formation ceases around the age of fifty years

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